Autoregressive LM

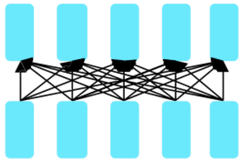
Wei Xu

(some slides from Greg Durrett)

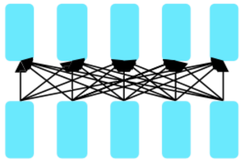
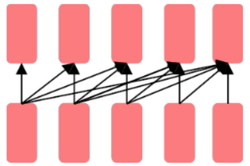
LM Pre-training

- Three types of neural architectures for LLM pre-training

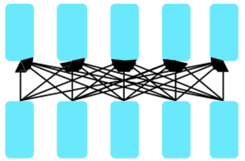
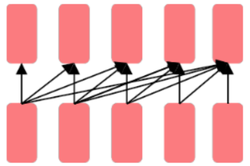
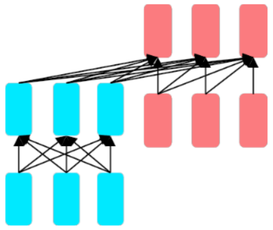
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    (e.g. GPT, Llama) → next-token prediction

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  - Encoder  bidirectional → masked token prediction
    (e.g., ELMo, BERT)
  - Decoder  unidirectional (autoregressive)
    (e.g. GPT, Llama) → next-token prediction
  - Encoder-Decoder 
    (e.g., BART, T5) encode bidirectionally
    → generate autoregressively

This Lecture

- Decoder-style LMs: GPT / GPT-2 / GPT-3
- In-context Learning
- Scaling Laws

GPT/GPT2/GPT3

OpenAI GPT/GPT2

- Train a single unidirectional transformer LM on long contexts

Radford et al. (2019)

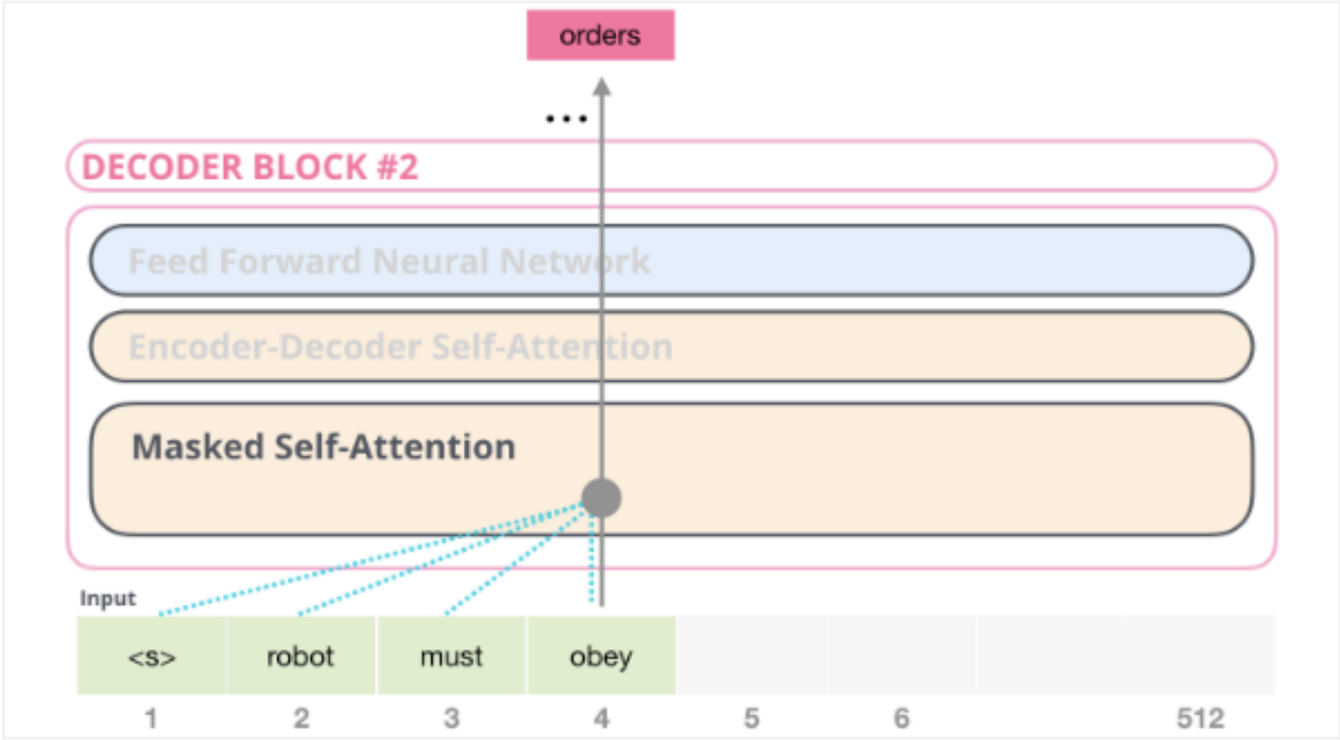
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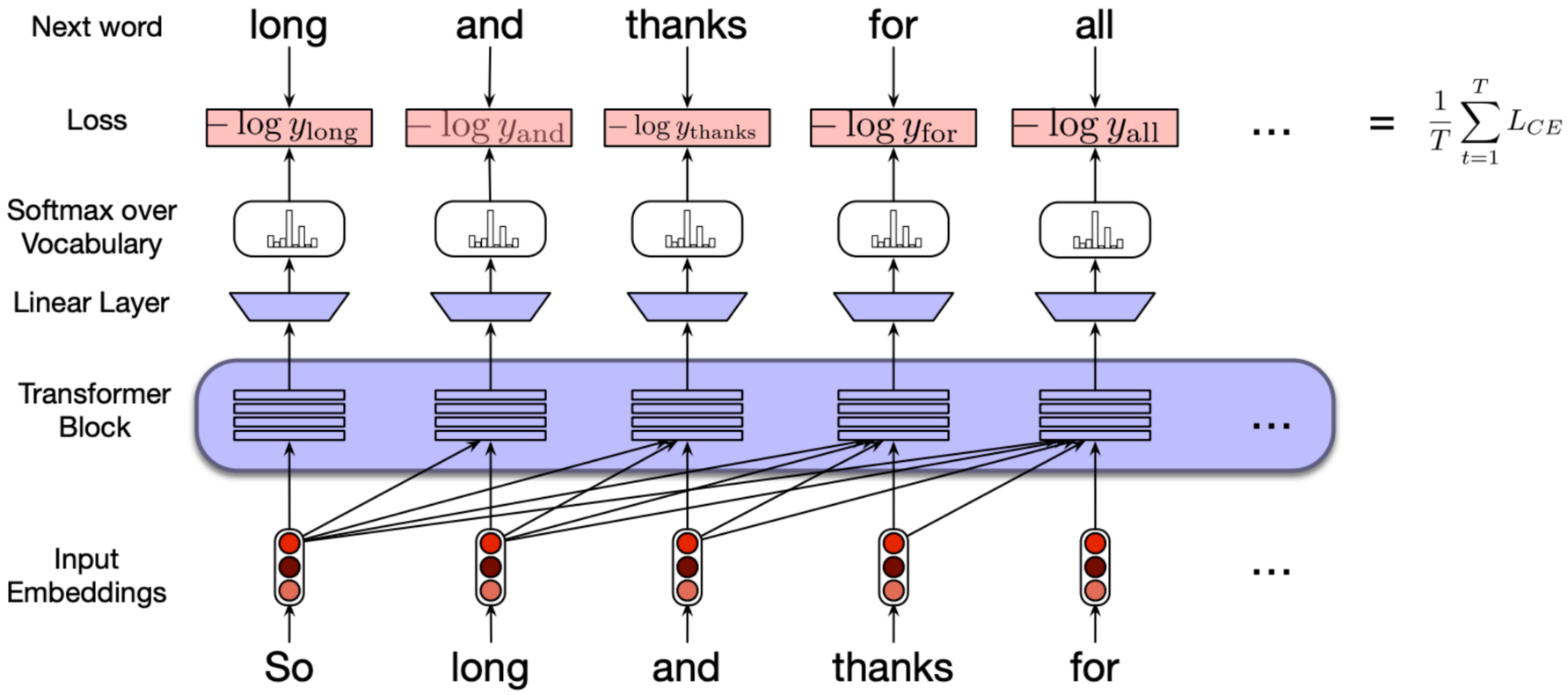


Image Credit: Jurafsky & Martin

OpenAI GPT/GPT2

- GPT2: trained on 40GB of text collected from upvoted links from reddit
- 1.5B parameters — the largest of these models trained as of March 2019

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

- Because it's a decoder-style language model, we can **generate** from it

Radford et al. (2019)

GPT-2

- Some results from GPT-2 foreshadowed the benefits of scaling up language models (shown here for Winograd and LM Perplexity)

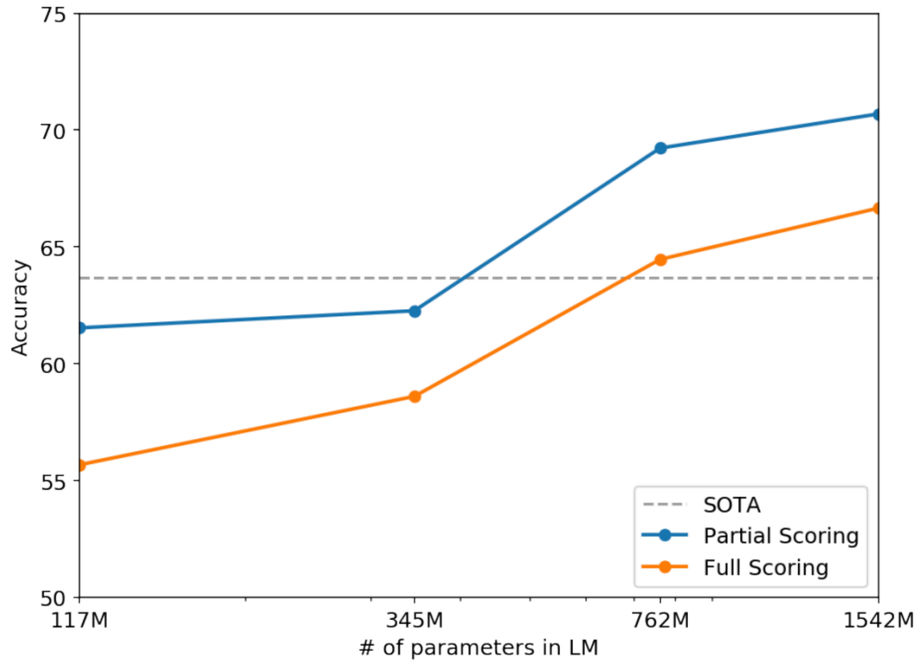


Figure 3. Performance on the Winogrand Schema Challenge as a function of model capacity.

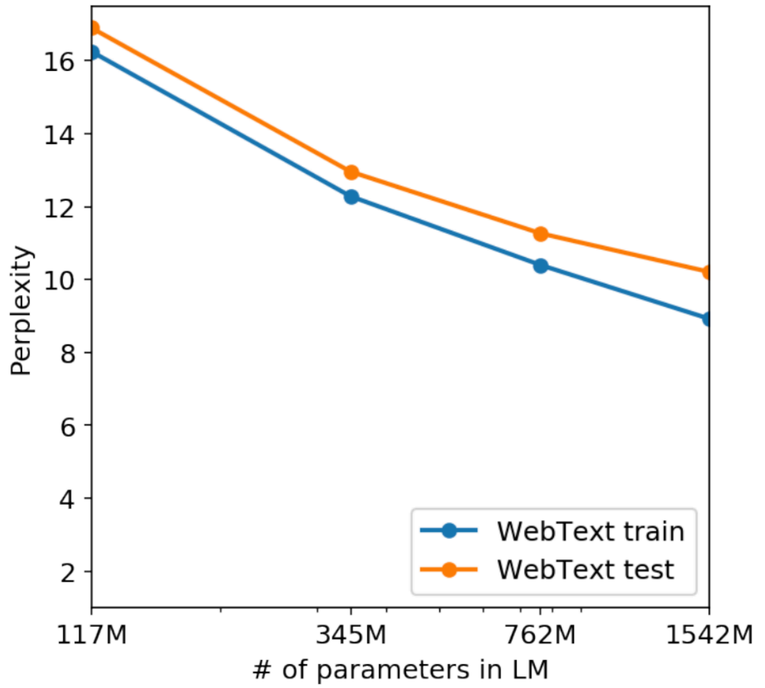
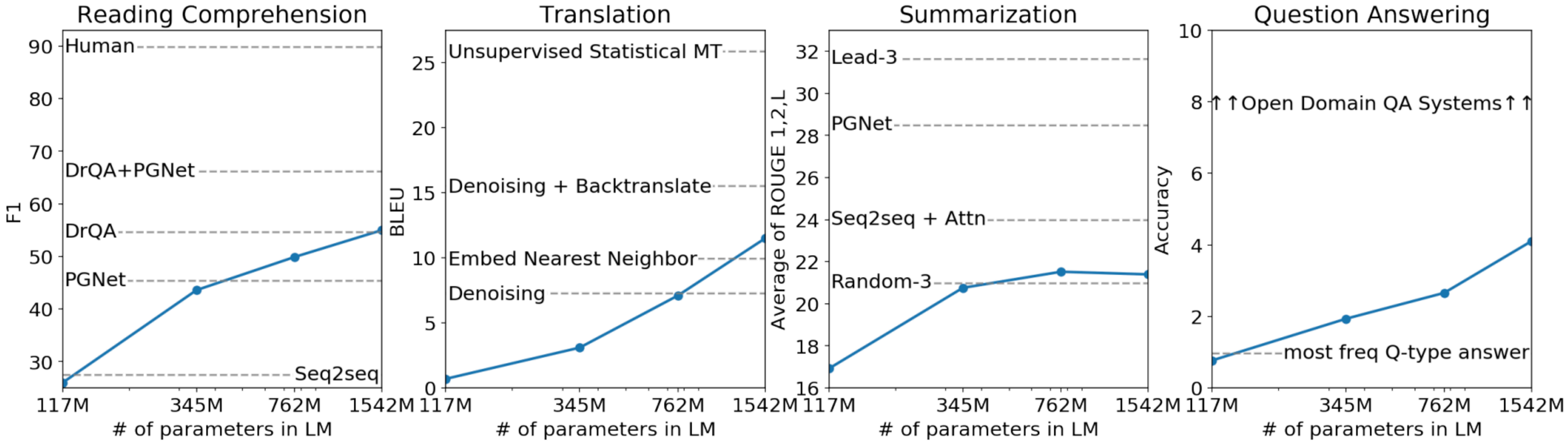


Figure 4. The performance of LMs trained on WebText as a function of model size.

Radford et al. (2019)

GPT-2

- Started to show strong zero-shot performance, even though “in terms of practical applications, the zero-shot performance of GPT-2 is still far from use-able.”



Radford et al. (2019)

OpenAI GPT2

System Prompt (human-written)

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

Model Completion (machine-written, 10 tries)

The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

Pérez and the others then ventured further into the valley. "By the time we reached the top of one peak, the water looked blue, with some crystals on top," said Pérez.

Pérez and his friends were astonished to see the unicorn herd. These creatures could be seen from the air without having to move

slide credit: OpenAI

OpenAI GPT2

System Prompt (human-written)

Miley Cyrus was caught shoplifting from Abercrombie and Fitch on Hollywood Boulevard today.

Model Completion (machine-written, 10 tries)

The 19-year-old singer was caught on camera being escorted out of the store by security guards.

The singer was wearing a black hoodie with the label 'Blurred Lines' on the front and 'Fashion Police' on the back.

Scroll down for video

Shoplifting: Miley Cyrus was caught shoplifting from Abercrombie and Fitch on Hollywood Boulevard today (pictured)

The singer was wearing a black hoodie with the label 'Blurred Lines' on the front and 'Fashion Police' on the back

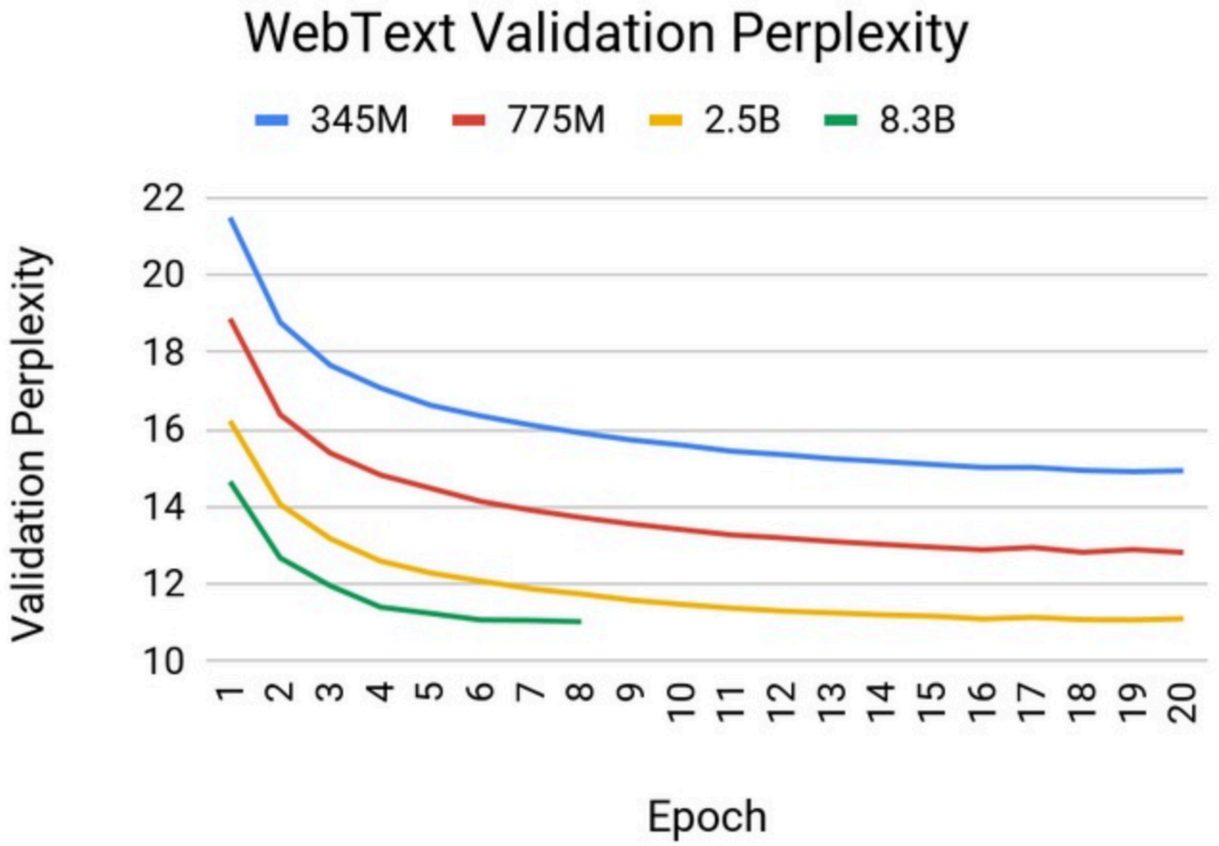
The singer was also wearing a pair of black-rimmed glasses, a black jacket, black jeans and black sandals.

She was carrying a pair of black and white striped gloves and a small black bag.

slide credit: OpenAI

GPT-3: Scaling Up

- Question: what are the scaling limits of large language models?
- NVIDIA: trained 8.3B parameter GPT model (5.6x the size of GPT-2), showed lower perplexity from this



GPT-3

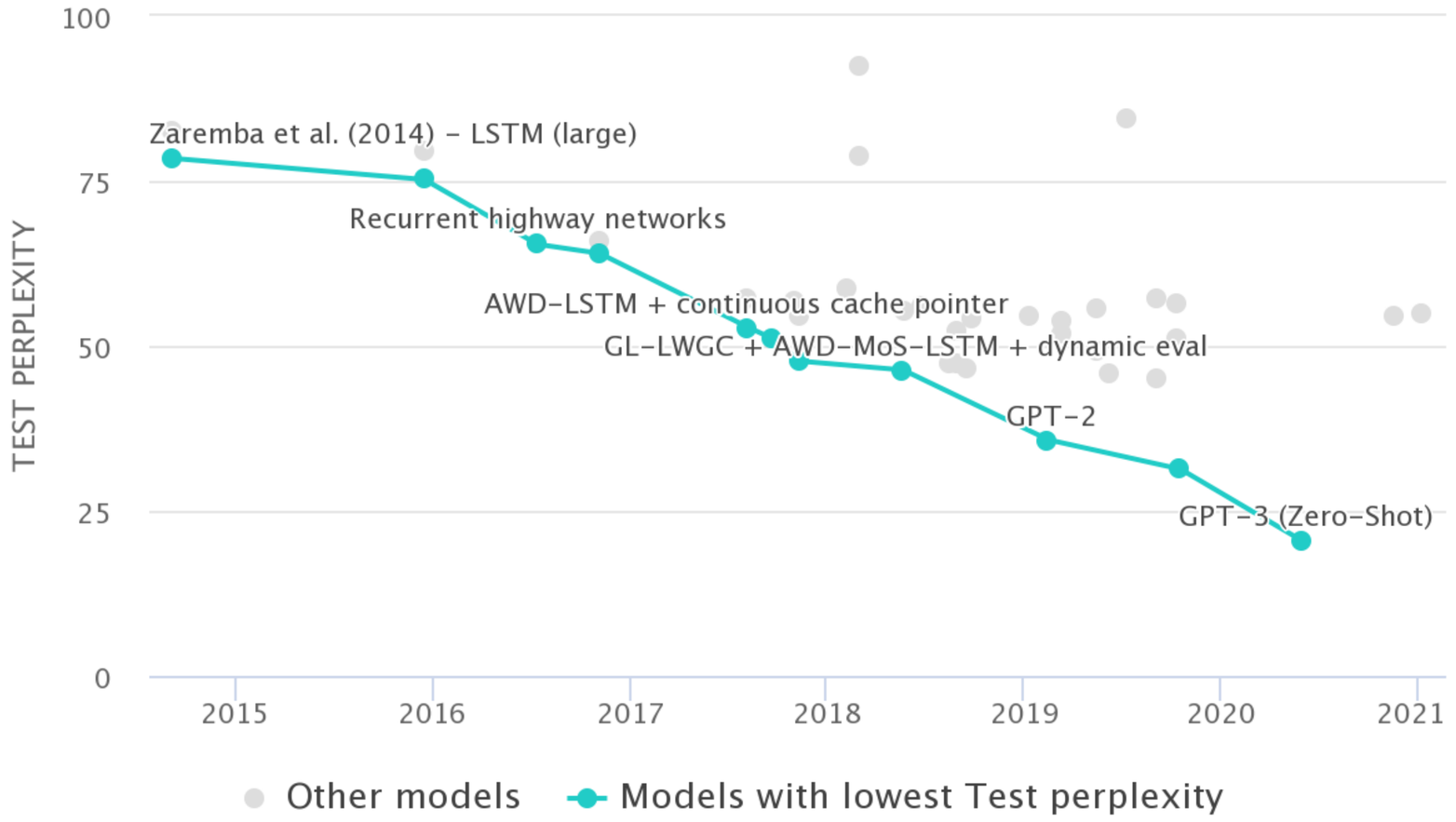
- GPT-2 but even larger: 1.3B (GPT2) —> 175B (GPT3) parameter models
- Sparse factorizations of the attention matrix to reduce computing time and memory use. context window is set to 2048 tokens.
- Data: filtered Common Crawl (410B tokens downsampled x0.44) + WebText dataset (19B x2.9) + two Internet-based book corpora (12Bx1.9, 55Bx0.43) + English Wiki (3B upsampled x3.4)

https://twitter.com/cocoweixu/status/1285727605568811011

Brown et al. (2020)

LM Evaluation - Perplexity

- The perplexity of modern language models have consistently been going down.



LM Evaluation - Perplexity

- Accuracy doesn't make sense — predicting the next word is generally impossible so accuracy values would be very low
- Evaluate LMs on the likelihood of held-out data (averaged to normalize for length)

$$\frac{1}{n}\sum_{i=1}^{n}\log P(w_i|w_1,\ldots,w_{i-1})$$



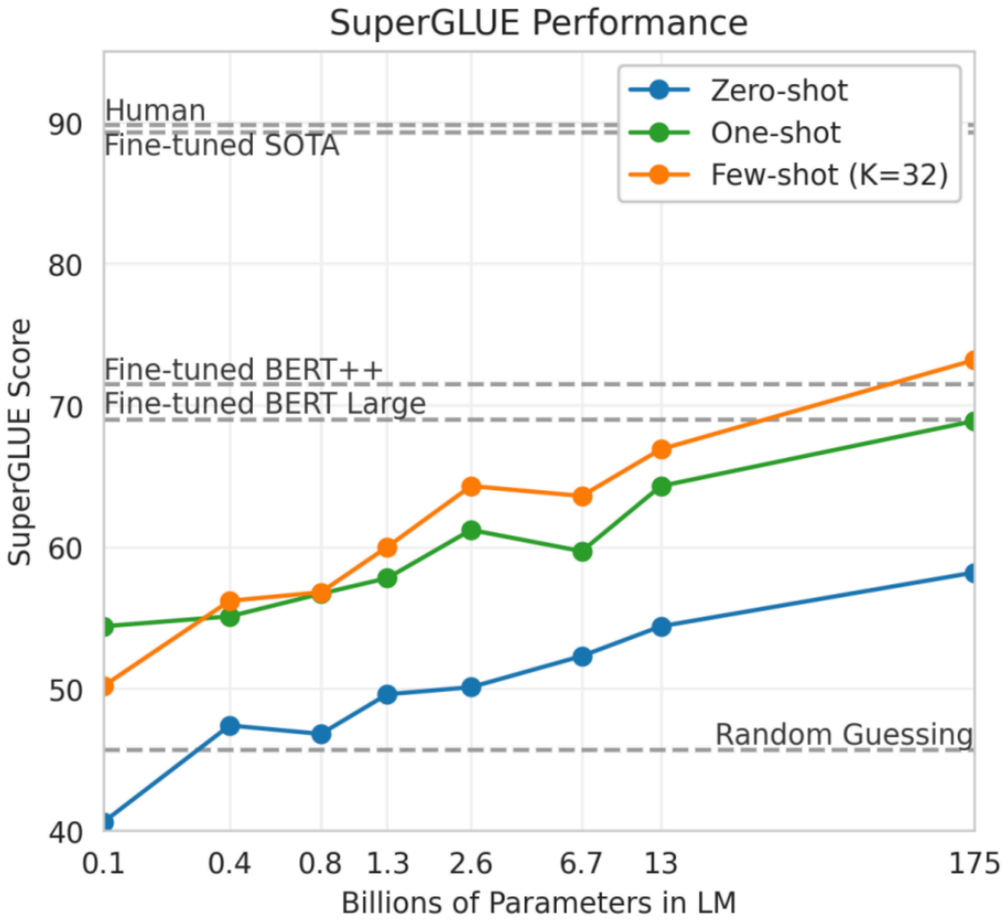
LM Evaluation - Perplexity

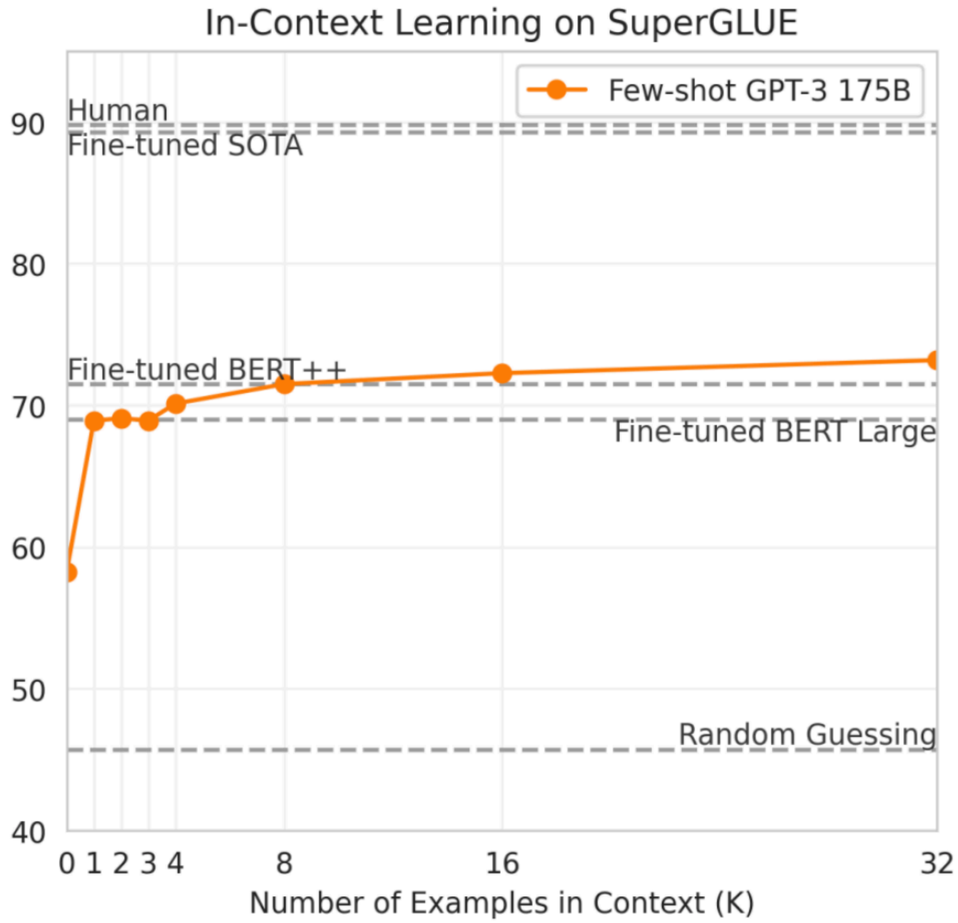
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- Evaluate LMs on the likelihood of held-out data (averaged to normalize for length)

$$\frac{1}{n}\sum_{i=1}^{n}\log P(w_i|w_1,\ldots,w_{i-1})$$

- Perplexity: exp(average negative log likelihood). Lower is better
  - Suppose we have probs 1/4, 1/3, 1/4, 1/3 for 4 predictions
  - Avg NLL (base e) = 1.242 Perplexity = 3.464 ← geometric mean of denominators

GPT-3 Performance





SuperGLUE benchmark - Wang et al. (2019)

SuperGLUE

Table 2: Development set examples from the tasks in SuperGLUE. **Bold** text represents part of the example format for each task. Text in *italics* is part of the model input. *Underlined* text is specially marked in the input. Text in a `monospaced font` represents the expected model output.

BoolQ	**Passage:** *Barq's – Barq's is an American soft drink. Its brand of root beer is notable for having caffeine. Barq's, created by Edward Barq and bottled since the turn of the 20th century, is owned by the Barq family but bottled by the Coca-Cola Company. It was known as Barq's Famous Olde Tyme Root Beer until 2012.* **Question:** *is barq's root beer a pepsi product* **Answer:** `No`
CB	**Text:** *B: And yet, uh, I we-, I hope to see employer based, you know, helping out. You know, child, uh, care centers at the place of employment and things like that, that will help out. A: Uh-huh. B: What do you think, do you think we are, setting a trend?* **Hypothesis:** *they are setting a trend* **Entailment:** `Unknown`
COPA	**Premise:** *My body cast a shadow over the grass.* **Question:** *What's the CAUSE for this?* **Alternative 1:** *The sun was rising.* **Alternative 2:** *The grass was cut.* **Correct Alternative:** `1`
MultiRC	**Paragraph:** *Susan wanted to have a birthday party. She called all of her friends. She has five friends. Her mom said that Susan can invite them all to the party. Her first friend could not go to the party because she was sick. Her second friend was going out of town. Her third friend was not so sure if her parents would let her. The fourth friend said maybe. The fifth friend could go to the party for sure. Susan was a little sad. On the day of the party, all five friends showed up. Each friend had a present for Susan. Susan was happy and sent each friend a thank you card the next week* **Question:** *Did Susan's sick friend recover?* **Candidate answers:** *Yes, she recovered* (`T`), *No* (`F`), *Yes* (`T`), *No, she didn't recover* (`F`), *Yes, she was at Susan's party* (`T`)
ReCoRD	**Paragraph:** *(CNN) Puerto Rico on Sunday overwhelmingly voted for statehood. But Congress, the only body that can approve new states, will ultimately decide whether the status of the US commonwealth changes. Ninety-seven percent of the votes in the nonbinding referendum favored statehood, an increase over the results of a 2012 referendum, official results from the State Electorcal Commission show. It was the fifth such vote on statehood. "Today, we the people of Puerto Rico are sending a strong and clear message to the US Congress ... and to the world ... claiming our equal rights as American citizens, Puerto Rico Gov. Ricardo Rossello said in a news release. @highlight Puerto Rico voted Sunday in favor of US statehood* **Query** For one, they can truthfully say, "Don't blame me, I didn't vote for them, " when discussing the <placeholder> presidency **Correct Entities:** `US`

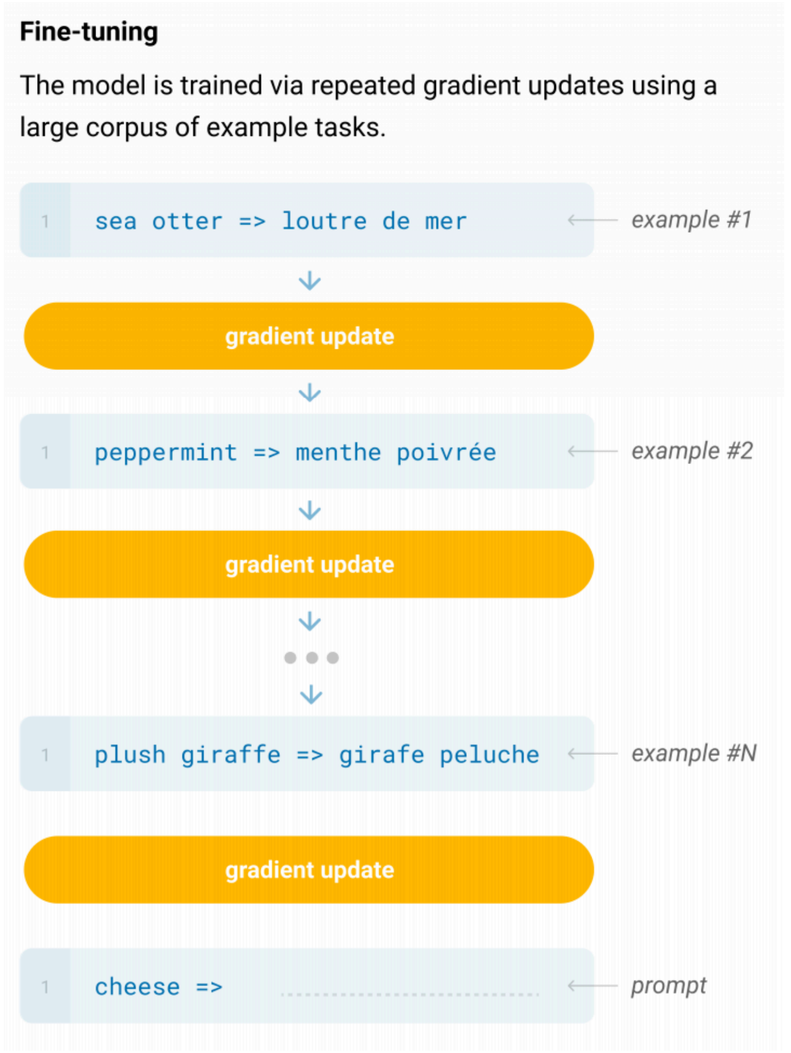
Wang et al. (2019)

Prompting

GPT-3: A Paradigm Shift!

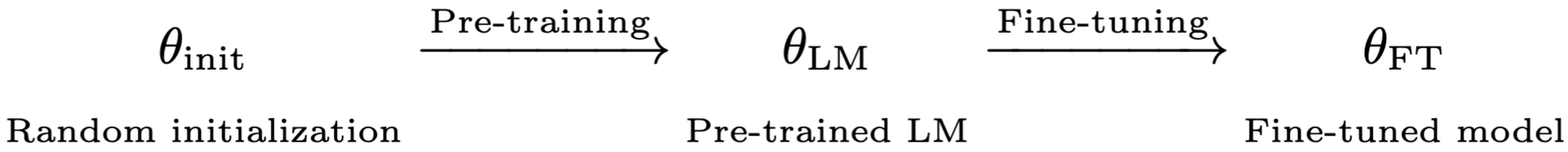
- Until GPT-3, fine-tuning on task-specific labeled datasets was the standard way to use pre-trained models (e.g., BERT, T5, GPT-2) for downstream tasks (e.g., text classification, NER, etc.)

- Fine-tuning requires computing gradients and updating model parameters — computationally expensive for a 175B GPT-3 model



Brown et al. (2020)

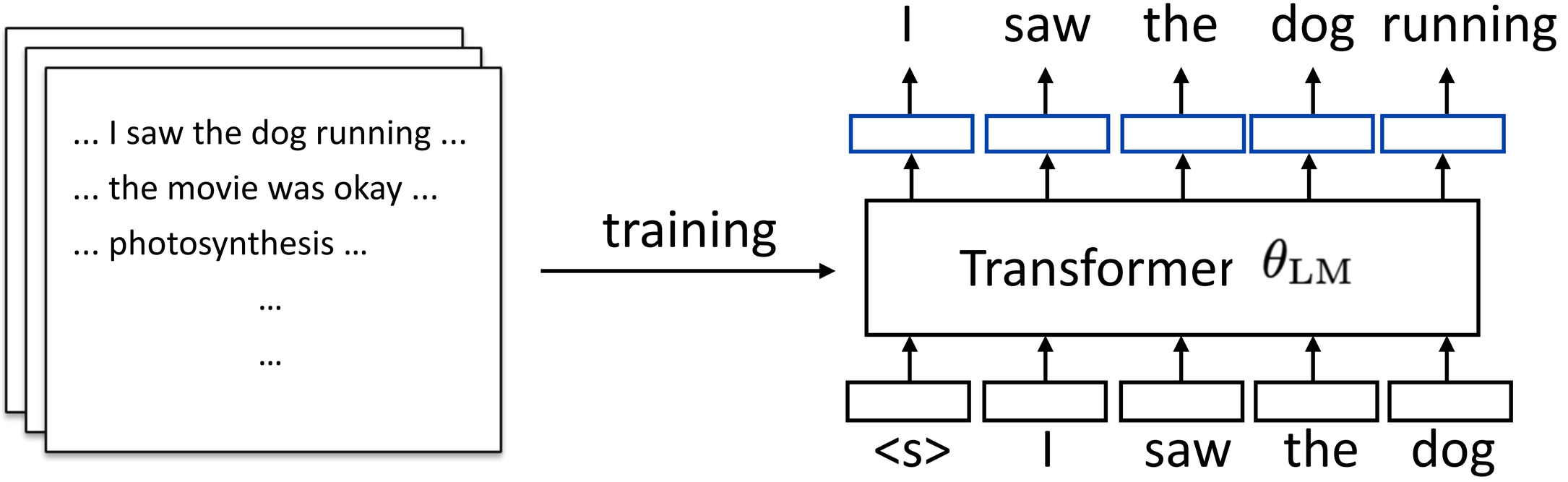
(Recap) Pre-training —> Fine-tuning Pipeline



Pretraining: Next-Token Prediction

- Autoregressive Language Models, i.e., next-token prediction:

$$\mathcal{L}_{\mathrm{NTP}}(\theta) = -\sum_{t=1}^{T} \log p_\theta(x_t \mid \mathbf{x}_{<t})$$



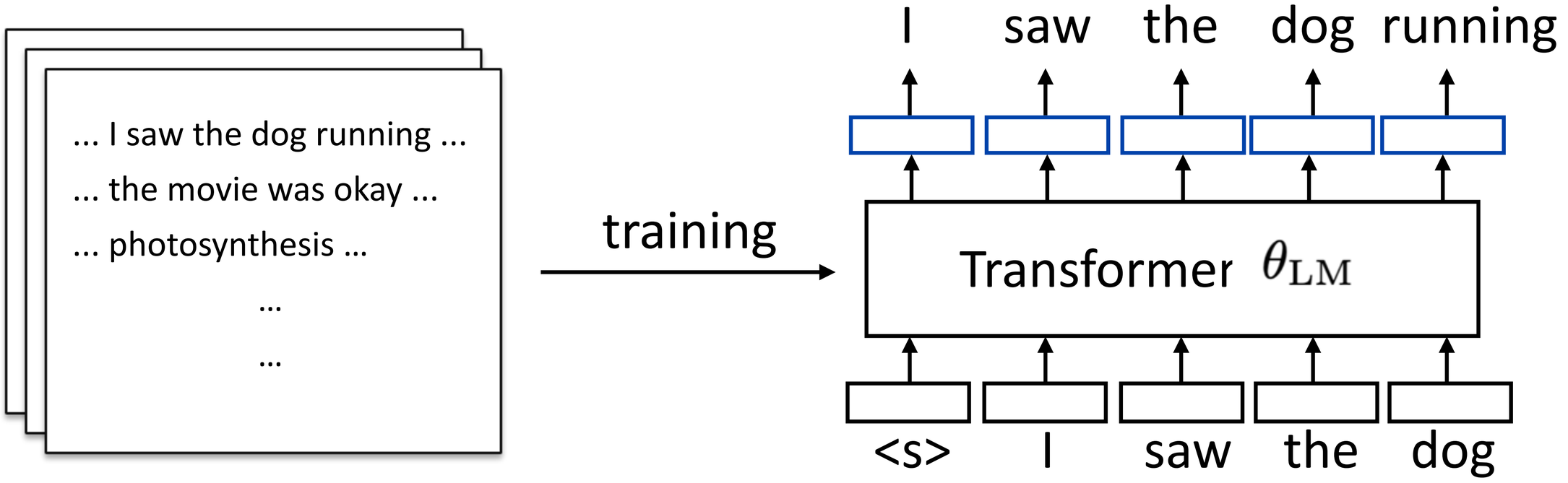
Large unlabeled corpus

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$x_1, x_2, \ldots, x_{t-1}$



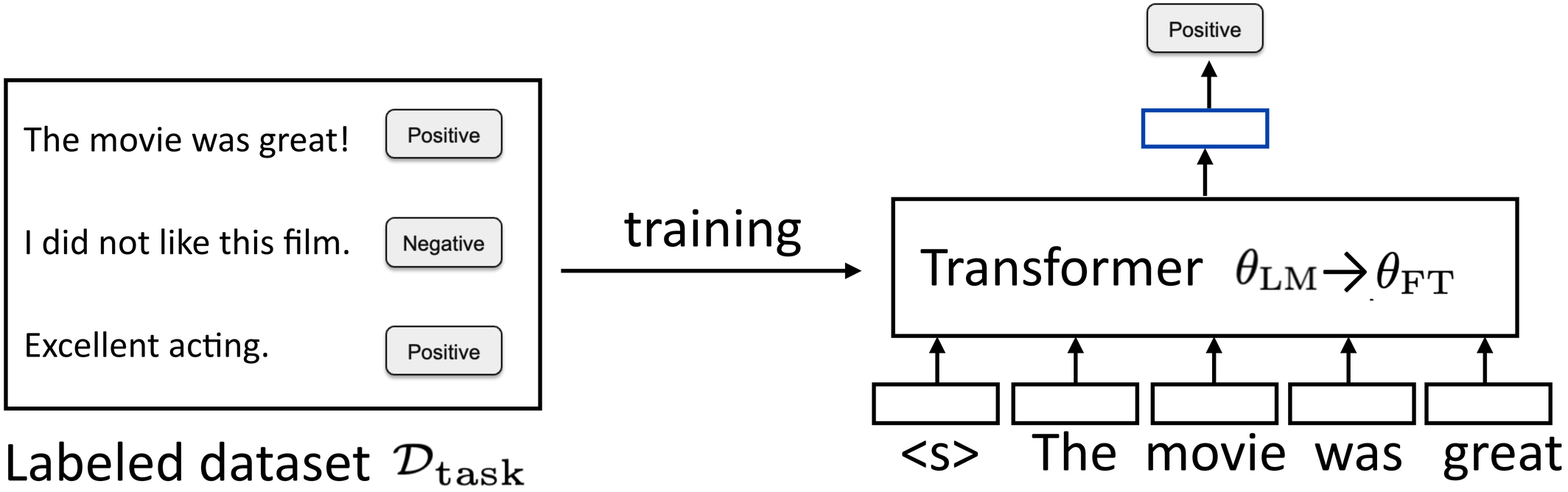
Large unlabeled corpus

Fine-tuning: Supervised Learning

- Initialize with pretrained LM weights; continue training on a labeled dataset (Compute loss → Backpropagation → Update model weights).

$$\mathcal{L}_{\text{finetune}}(\theta) = -\sum_{(\mathbf{x},\mathbf{y})\in\mathcal{D}_{\text{task}}} \log p_\theta(\mathbf{y} \mid \mathbf{x})$$

initialize from θ_{LM}

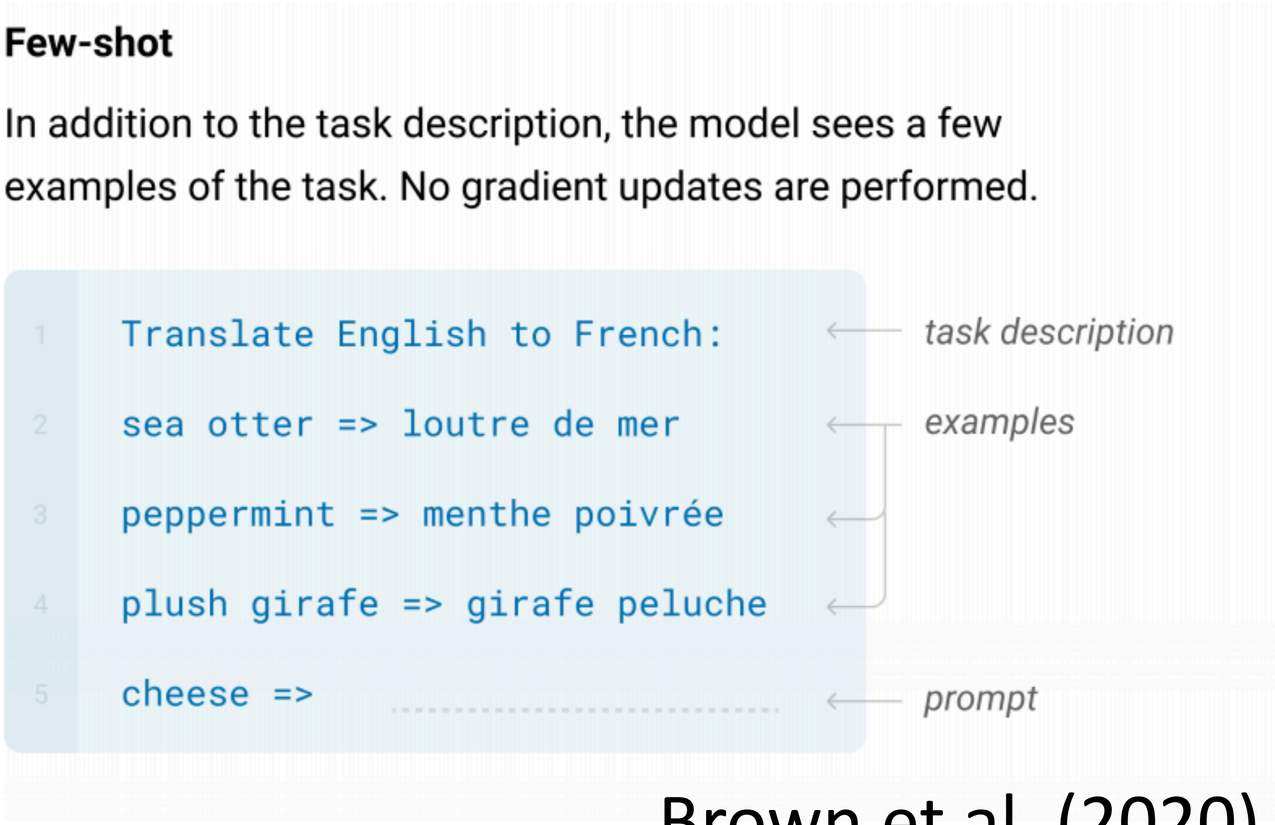


GPT-3: Few-shot Prompting

- Model is frozen. No parameter updates.
  (i.e., only a forward pass, no backward pass)

- Prompt: task instruction plus a few demonstrations (training examples)

- This is also called:
  in-context learning



Brown et al. (2020)

GPT-3: Few-shot Prompting

- Model is frozen and is given a few demonstrations.

Circulation revenue has increased by 5% in Finland. // Positive

Panostaja did not disclose the purchase price. // Neutral

Paying off the national debt will be extremely painful. // Negative

The company anticipated its operating profit to improve. // _____

Circulation revenue has increased by 5% in Finland. // Finance

They defeated … in the NFC Championship Game. // Sports

Apple … development of in-house chips. // Tech

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Brown et al. (2020), Schick et al. (2020), Schick and Schütze (2021)

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- "in-context learning" - unlike conventional machine learning in that there's no optimization of any parameters.

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- "in-context learning" - unlike conventional machine learning in that there's no optimization of any parameters.
- Model "learns" by conditioning on a few examples of the task.

Brown et al. (2020), Schick et al. (2020), Schick and Schütze (2021)

Prompting

- Zero-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\mathrm{LM}}}(\mathbf{y} \mid \mathbf{i}, \mathbf{x})$$

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:        ← i, task description
2  cheese =>                           ← x
```

Brown et al. (2020)

Prompting

- Few-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\text{LM}}} \left(\mathbf{y} \mid \mathbf{i}, \underbrace{(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_K, \mathbf{y}_K)}_{\text{Few-shot examples}}, \mathbf{x} \right)$$

- Zero-shot Prompt:

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The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:
2  cheese =>
```

$\mathbf{i}$ *task description*

$\mathbf{x}$

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:
2  sea otter => loutre de mer
3  peppermint => menthe poivrée
4  plush girafe => girafe peluche
5  cheese =>
```

$\mathbf{i}$ *task description*

examples $(\mathbf{x}_1, \mathbf{y}_1)$ ⋮

$\mathbf{x}$

Brown et al. (2020)

Prompting: Inference Without Training

- Few-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\mathrm{LM}}} \left(\mathbf{y} \mid \mathbf{i}, \underbrace{(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_K, \mathbf{y}_K)}_{\text{Few-shot examples}}, \mathbf{x} \right)$$

- Zero-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\mathrm{LM}}} (\mathbf{y} \mid \mathbf{i}, \mathbf{x})$$

- A pretrained LM performs new tasks by conditioning on instructions, (optional) examples, input text.
- No backward pass. No gradients. No parameter updates (parameters of LM are frozen).

Brown et al. (2020)

GPT-3

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	**89.0**	**91.0**	**96.9**	**93.9**	**94.8**	**92.5**
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0

	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	**76.1**	**93.8**	**62.3**	**88.2**	**92.5**	**93.3**
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

- Sometimes very impressive, (MultiRC, ReCoRD), sometimes very bad
- Results on other datasets are equally mixed — but still strong for a few-shot model!

Brown et al. (2020)

MultiRC (multi-sentence)

Sent 1: The hijackers attacked at 9:28.
Sent 2: While traveling 35,000 feet above eastern Ohio, United 93 suddenly dropped 700 feet.
Sent 3: Eleven seconds into the descent, the FAA's air traffic control center in Cleveland received the first of two radio transmissions from the aircraft.
Sent 4: During the first broadcast, the captain or first officer could be heard declaring "Mayday" amid the sounds of a physical struggle in the cockpit.
Sent 5: The second radio transmission, 35 seconds later, indicated that the fight was continuing.
Sent 6: The captain or first officer could be heard shouting: "Hey get out of here-get out of here-get out of here."
Sent 7: On the morning of 9/11, there were only 37 passengers on United 93-33 in addition to the 4 hijackers.
Sent 8: This was below the norm for Tuesday mornings during the summer of 2001.
Sent 9: But there is no evidence that the hijackers manipulated passenger levels or purchased additional seats to facilitate their operation.
Sent 10: The terrorists who hijacked three other commercial flights on 9/11 operated in five-man teams.
Sent 11: They initiated their cockpit takeover within 30 minutes of takeoff.
Sent 12: On Flight 93, however, the takeover took place 46 minutes after takeoff and there were only four hijackers.

Question: Which two factors were different between the three other hijacked planes and United 93?
the day of the takeover
A) The amount of time that passed before the takeover started
B)* United 93 took longer and had less hijackers
C) The airline operating the planes
D) The weather and fuel used by the airplane
E) The navigation system used by the planes
Reasoning needed: Discourse relation (contrast)
One needs to identify that the discourse marker *however* in Sent 12 indicates a contrast relation between Flight 93 and the flights mentioned in Sent 10. Also, *only* in Sent 12 indicates that the number of hijackers were fewer than in the contrasted other flights.

Question: What was below average for this particular day?
A) the number of passengers in the first class.
B)* the number of passengers on board.
C) the number of hijackers
D) the amount of air traffic in the skies
E) the temperature
Reasoning needed: Event coreference
One needs to identify that *This* in Sent 8 co-refers to (event of) number of passengers in Sent 7. Note that Sent 12 contains *only four hijackers* and understanding that *only* indicates a smaller number of entities than expected (as in previous question), might mislead a system into believing that (C) is the correct answer.

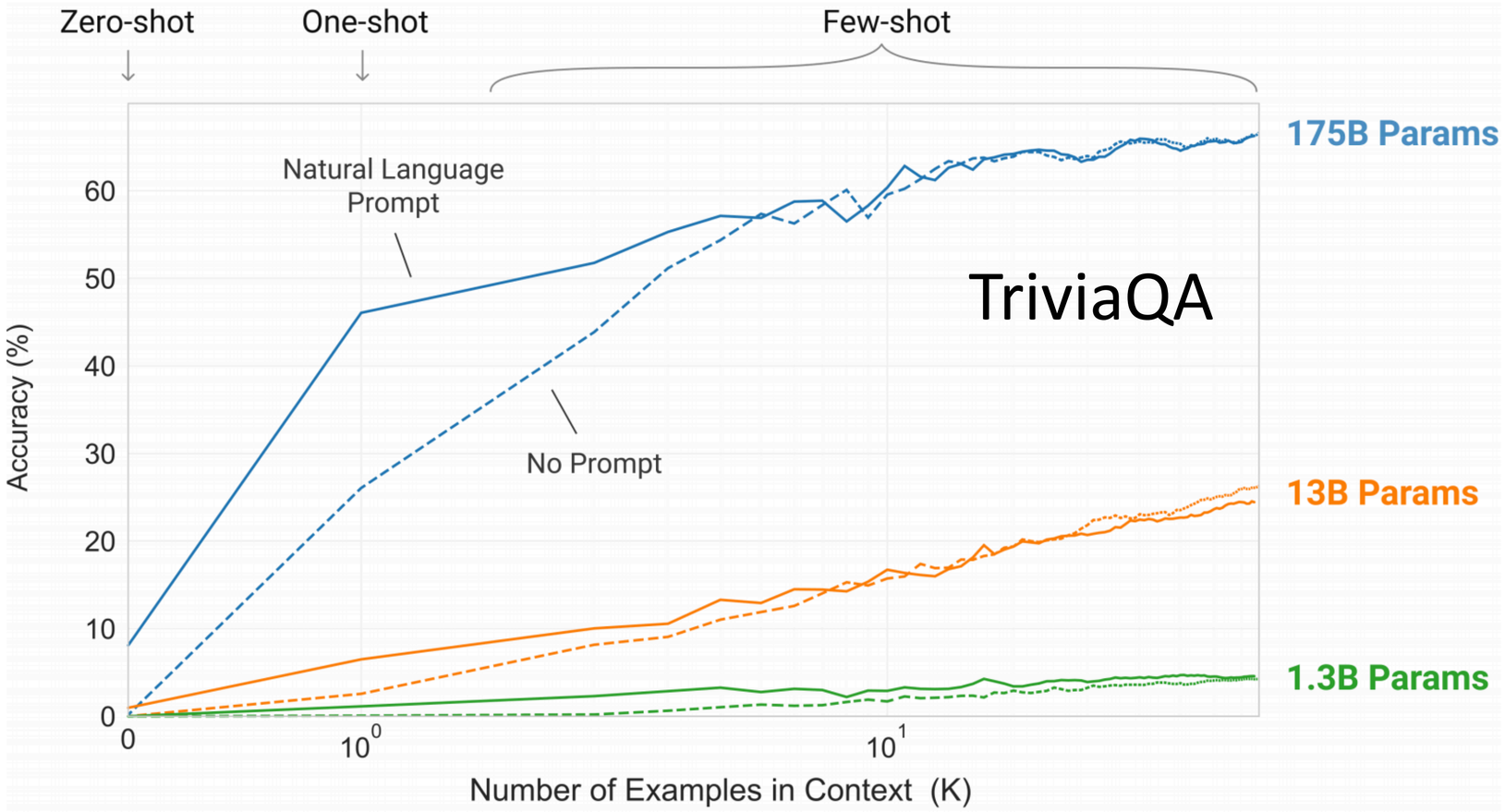
TriviaQA

Context →	Q: ‘Nude Descending A Staircase’ is perhaps the most famous painting by which 20th century artist? A:
Target Completion →	MARCEL DUCHAMP
Target Completion →	r mutt
Target Completion →	duchamp
Target Completion →	marcel duchamp
Target Completion →	R.Mutt
Target Completion →	Marcel duChamp
Target Completion →	Henri-Robert-Marcel Duchamp
Target Completion →	Marcel du Champ
Target Completion →	henri robert marcel duchamp
Target Completion →	Duchampian
Target Completion →	Duchamp
Target Completion →	duchampian
Target Completion →	marcel du champ
Target Completion →	Marcel Duchamp
Target Completion →	MARCEL DUCHAMP

Figure G.34: Formatted dataset example for TriviaQA. TriviaQA allows for multiple valid completions.

GPT-3: Few-shot Learning

- **Key observation:** few-shot learning only works with the very largest models!
- **Later (2022-present):** smaller models also can do this now, by training longer on larger data, instruction-tuning, and post-training,



Brown et al. (2020)

GPT-3

- GPT-2 but even larger: 1.3B —> 175B parameter models

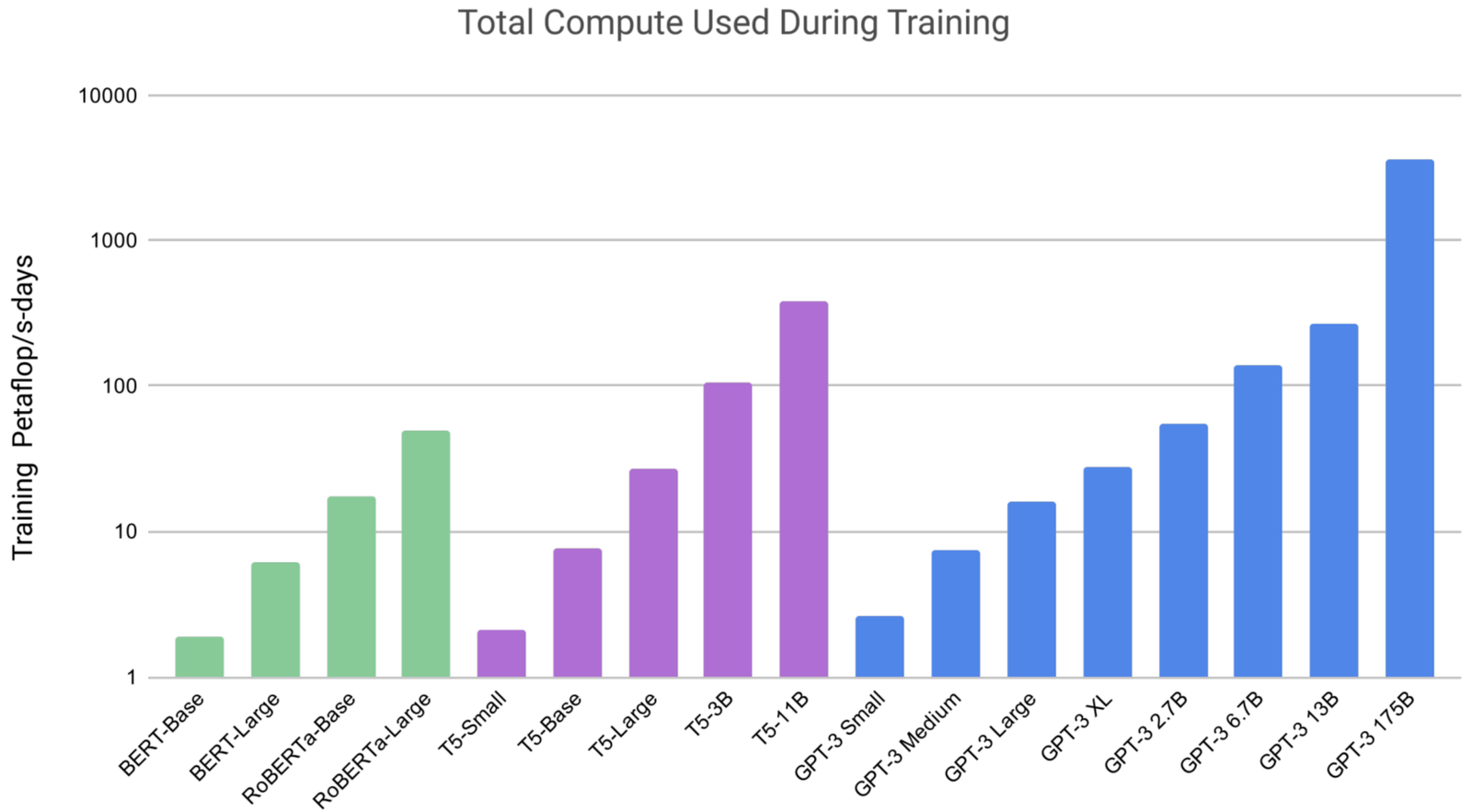
Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

- Trained on 570GB of Common Crawl
- 175B parameter model’s parameters alone take >400GB to store (4 bytes per param). Trained in parallel on a “high bandwidth cluster provided by Microsoft”

Brown et al. (2020)

GPT-3 Pre-training Cost

- Trained on Microsoft Azure, estimated to cost $4~10M (1000x BERT-large)

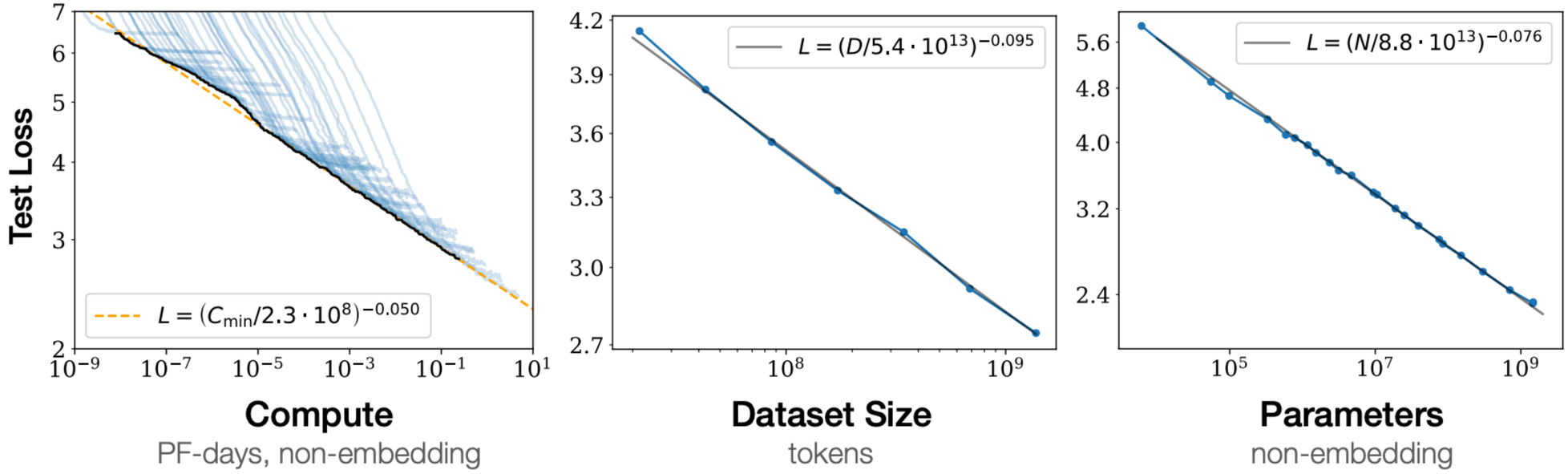


1 petaflop/s-day is equivalent to 8 V100 GPUs at full efficiency of a day

Brown et al. (2020)

GPT Scaling Laws

- With more compute, larger models get further down the loss “frontier”
- Building a bigger model (increasing compute) will decrease test loss!



petaflop (10^{20})/s-days

Kaplan et al. (2020)

Scaling Laws

Wei Xu

Compute Cost

- For a fixed compute budget C, there is a tradeoff between N and D

- For example, training compute for a Transformer LM is approximately:

$$C \approx 6ND$$

$$(\text{FLOPs} \approx 6 \cdot \text{params} \cdot \text{tokens})$$

  - Model size **N**: number of parameters
  - Training dataset size **D**: number of tokens/examples
  - Training compute **C**: floating point operations (FLOPs)

Training LLMs under Resource Limits

- Typically, there is:
  - a limit on compute budget (number of GPUs × training time)
  - a limit on the amount of training data available
- To build a very large language model, we can only afford training once
- Q: What is the optimal model size (number of parameters) to train?

Scaling Laws

- How do we allocate the resources more optimally?
- → How model performance would change as resources increase?

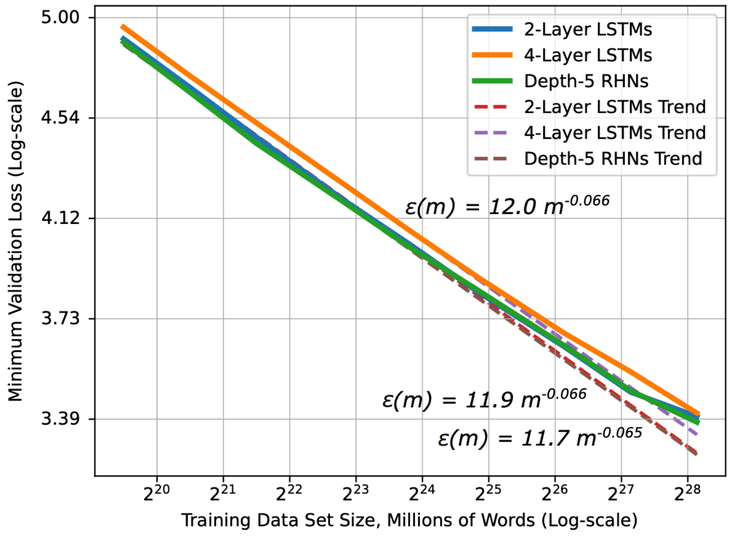
Performance = function (N, D, C)

  - Model size **N**: number of parameters
  - Training dataset size **D**: number of tokens/examples
  - Training compute **C**: floating point operations (FLOPs)

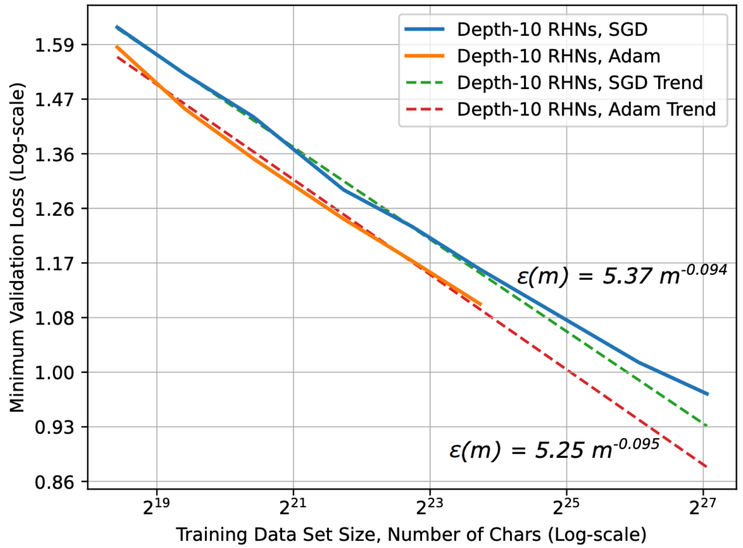
- Other factors (e.g., model architecture, hyperparameters) fixed/tuned

Baidu Scaling Laws

- Predates modern LLM scaling laws
- **Power-law** relationships with dataset and model size
- Observed across diverse tasks and neural network architectures



Machine Translation
(seq2seq - Luong et a. 2015)



Character Language Models
(RHN - Zilly et al. 2017)

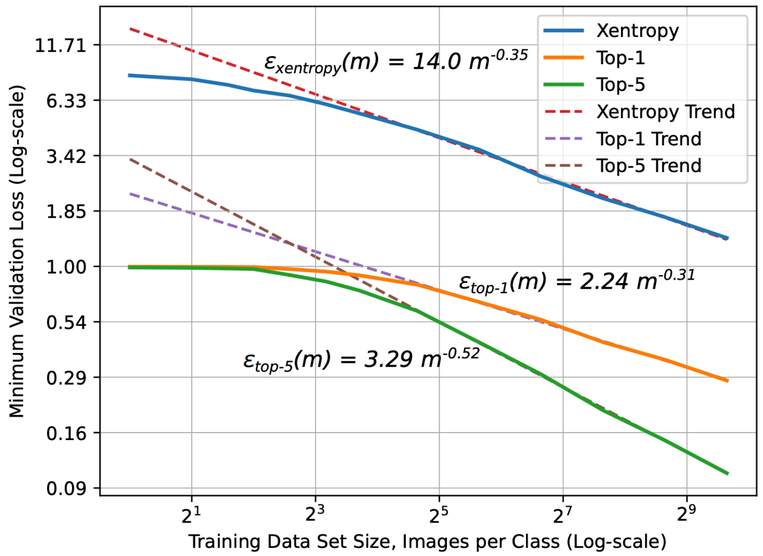
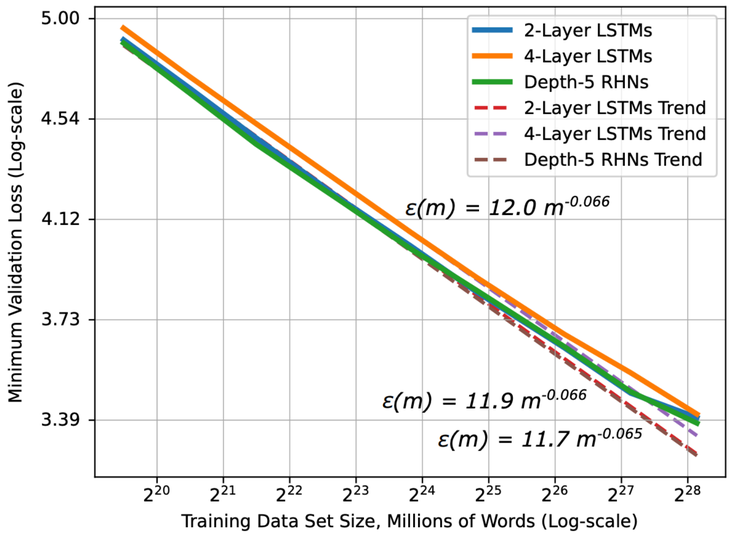


Image Classification
(ResNets - He et al. 2016)

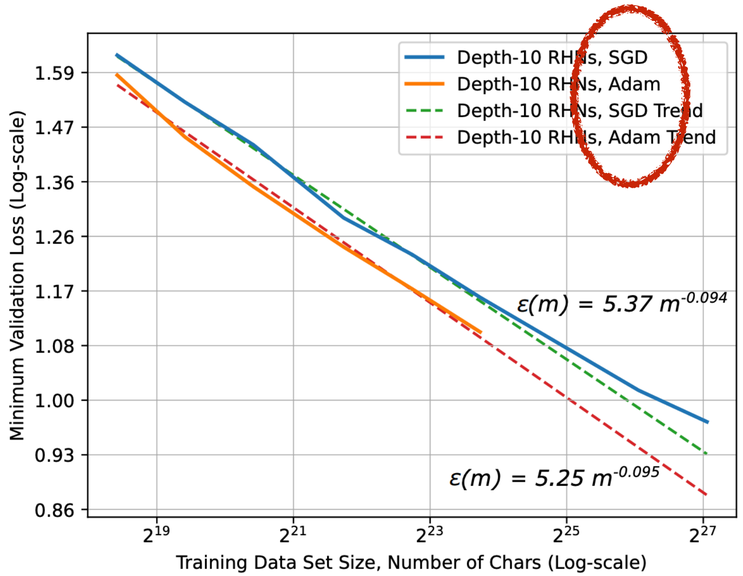
Hestness et al. (2017)

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- Predates modern LLM scaling laws
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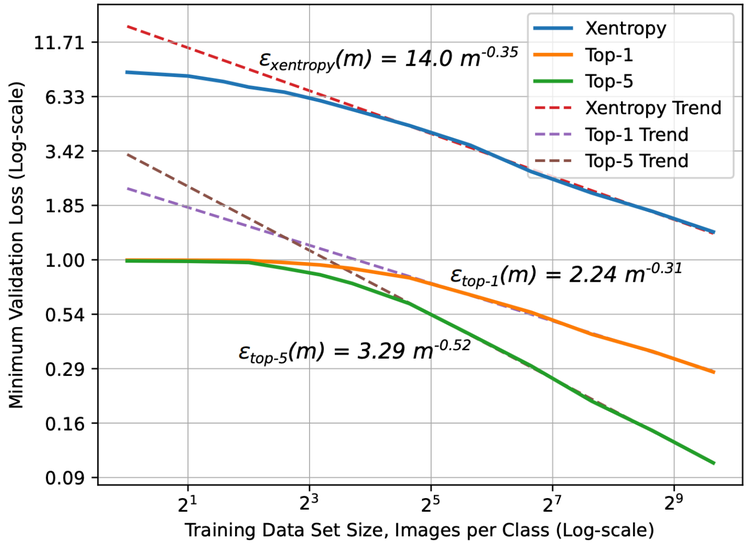
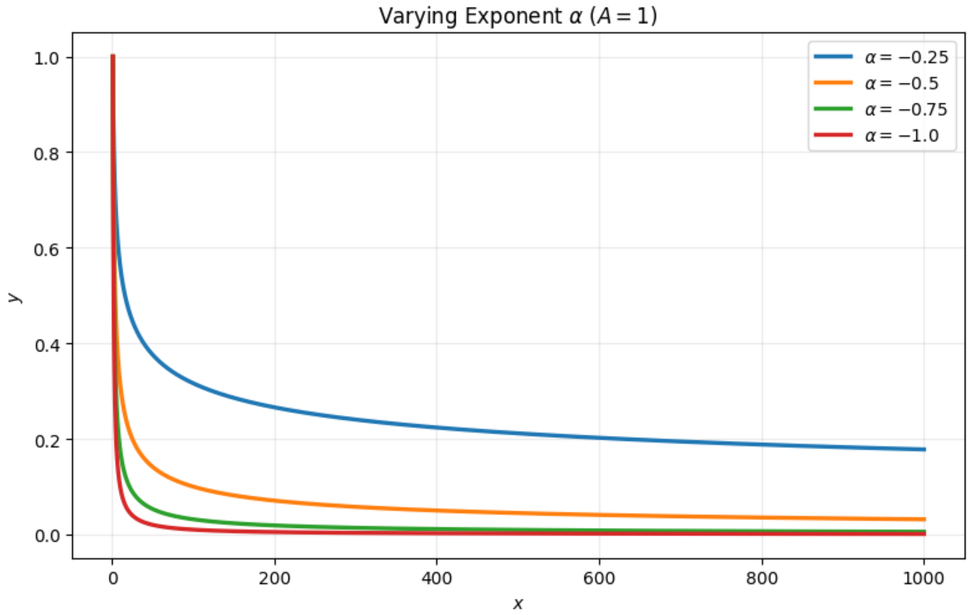


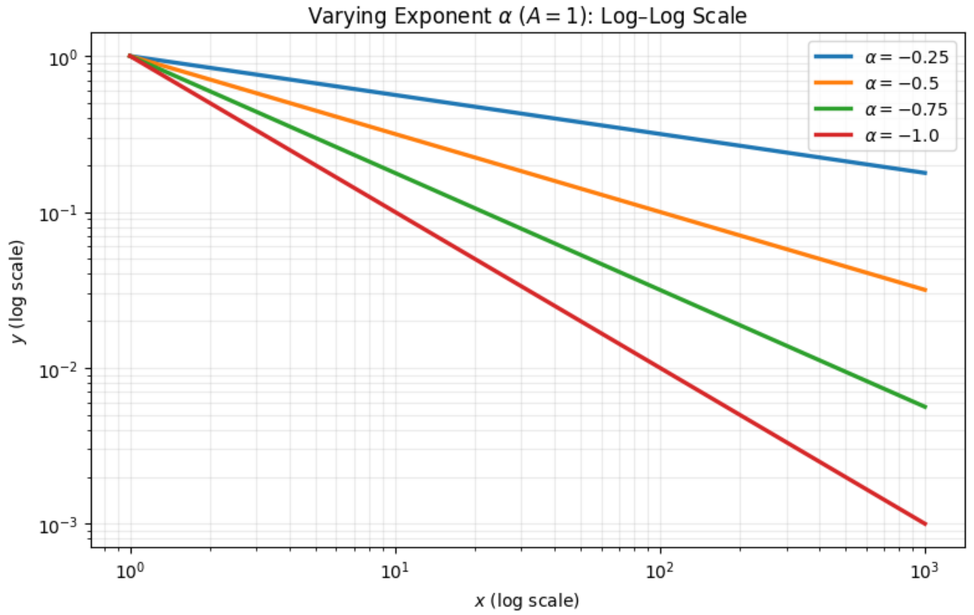
Image Classification
(ResNets - He et al. 2016)

Hestness et al. (2017)

Power Law

$$y = Ax^{\alpha} \quad \Longrightarrow \quad \log y = \underbrace{\log A}_{\text{intercept}} + \underbrace{\alpha}_{\text{slope}} \log x$$

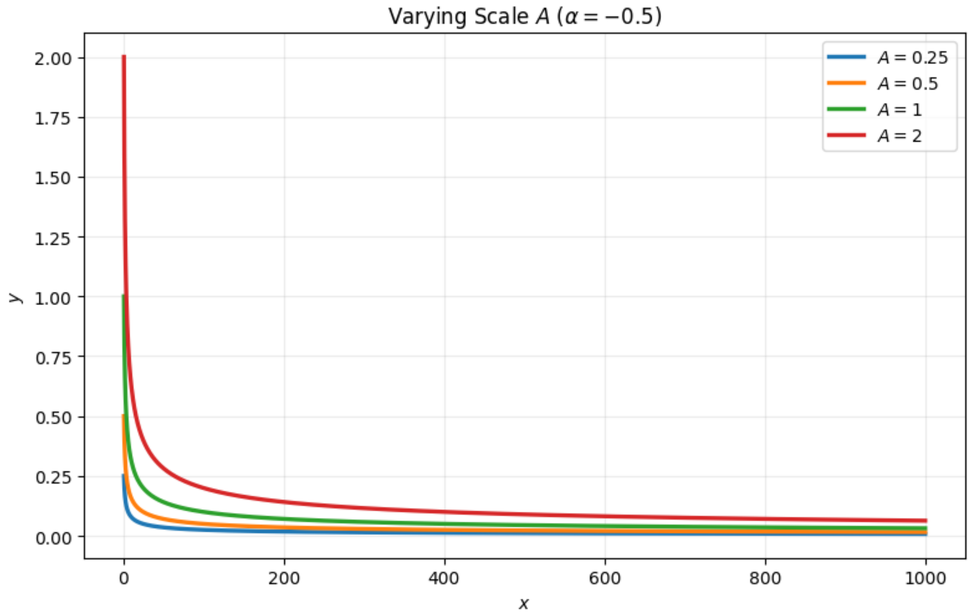


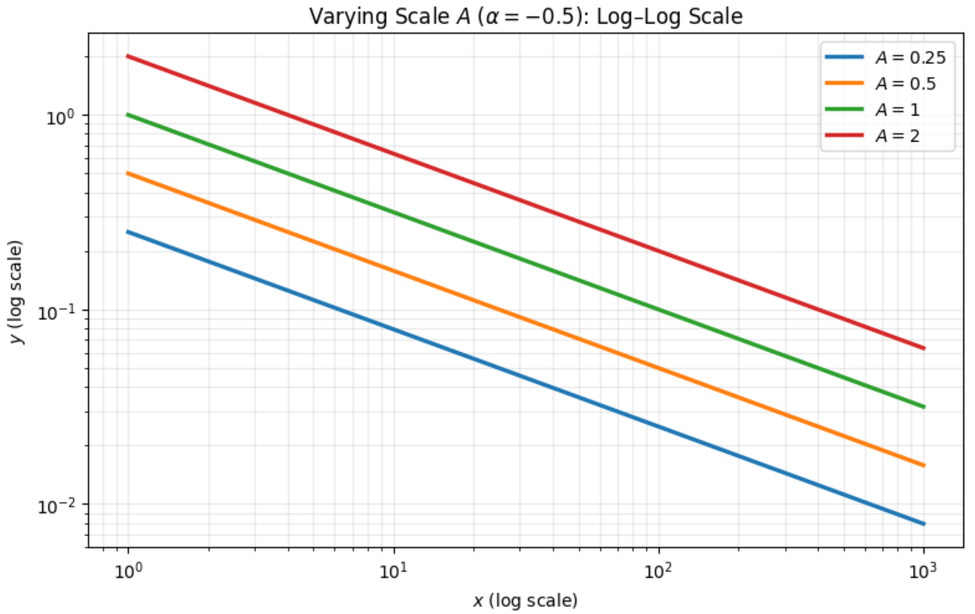


- More negative α means faster improvement (lower loss/error) with scale

Power Law

$$y = Ax^{\alpha} \quad \Longrightarrow \quad \log y = \underbrace{\log A}_{\text{intercept}} + \underbrace{\alpha}_{\text{slope}} \log x$$





- From Baidu's scaling law paper: improved model architectures and optimizers can improve the power-law intercept, but not the exponent
- i.e., models for a single domain show the same learning curve steepness.

- Is power-law growth the same as exponential growth?

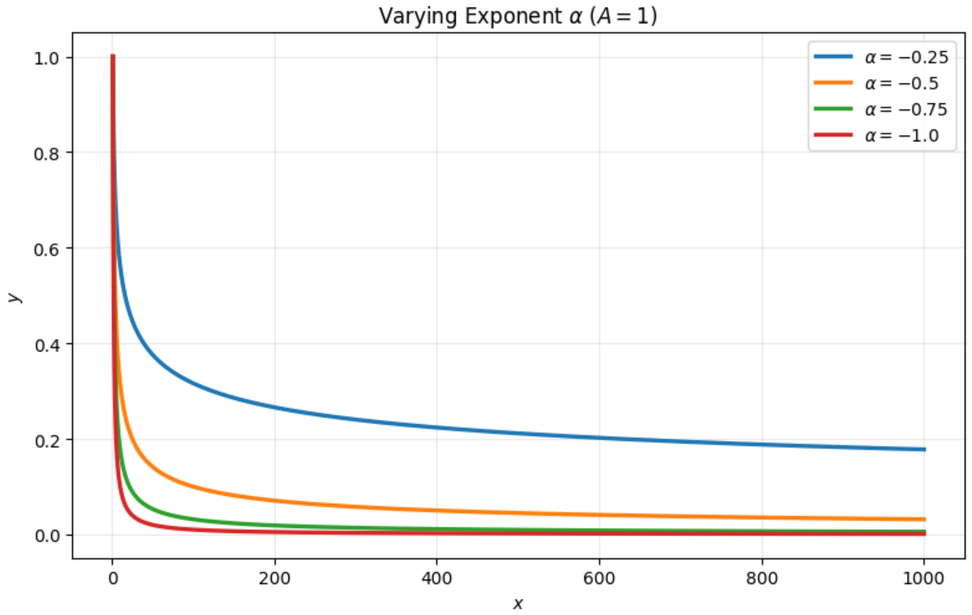


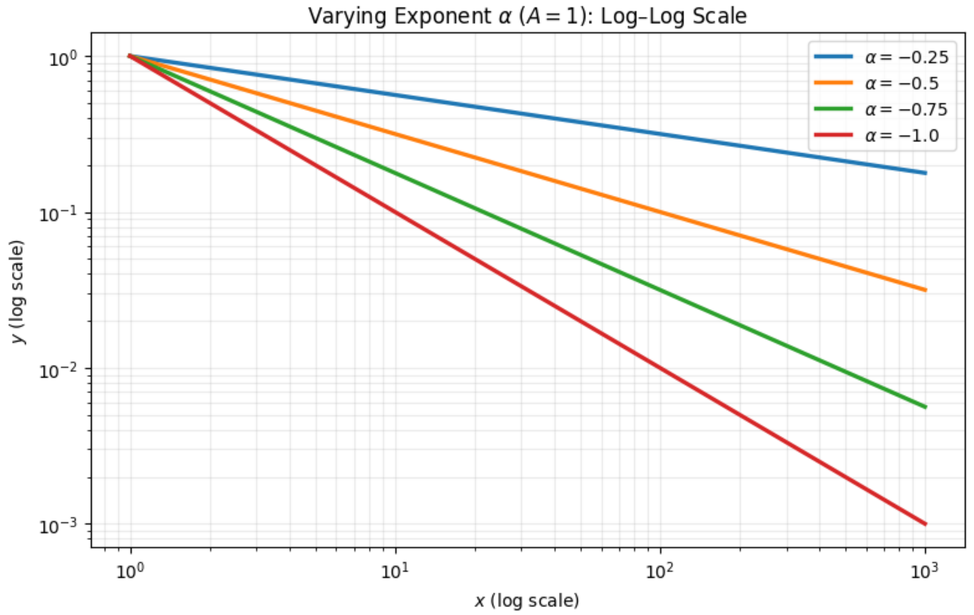
Power Law vs. Exponential

- Power law: $y = Ax^{\alpha} \implies \log y = \log A + \alpha \log x$
- Exponential: $y = Ae^{kx} \implies \log y = \log A + kx$

Power Law

$$y = Ax^{\alpha} \quad \Longrightarrow \quad \log y = \underbrace{\log A}_{\text{intercept}} + \underbrace{\alpha}_{\text{slope}} \log x$$

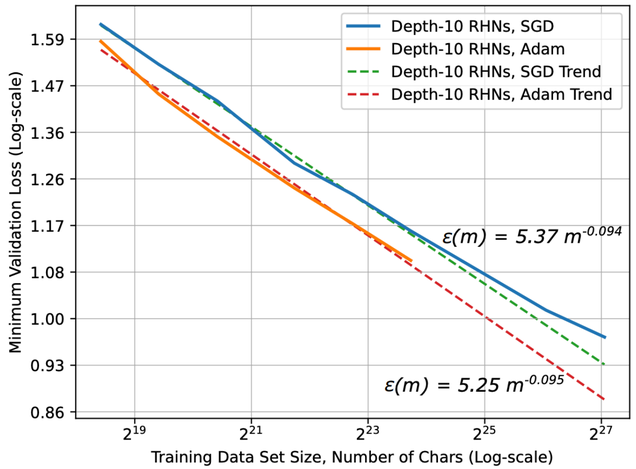




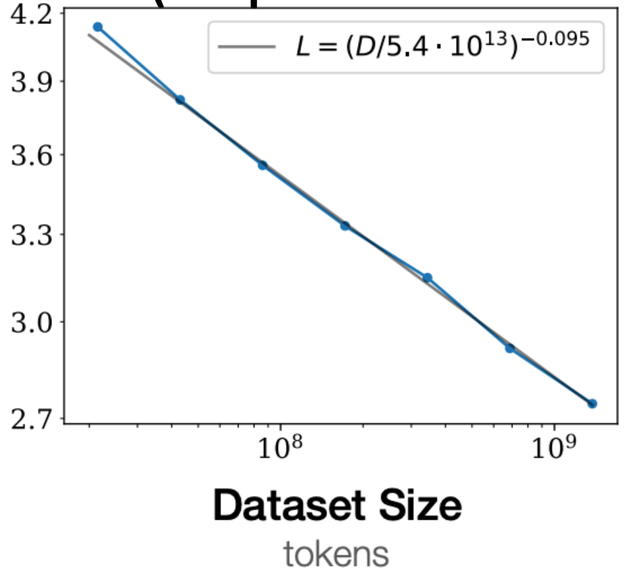
- More negative α means faster improvement (lower loss/error) with scale

Dataset Scaling Laws

Baidu (Hestness et al. 2017)

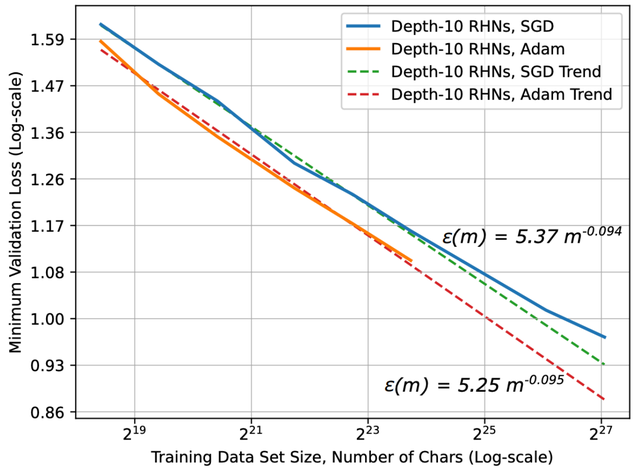


OpenAI (Kaplan et al. 2020)



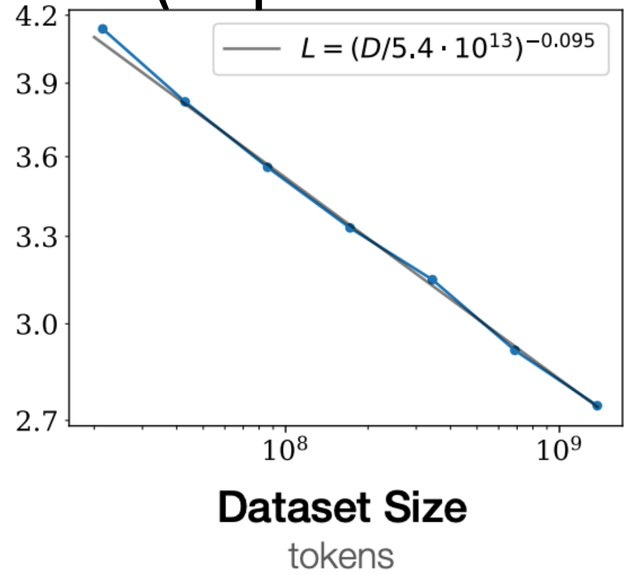
Dataset Scaling Laws

Baidu (Hestness et al. 2017)



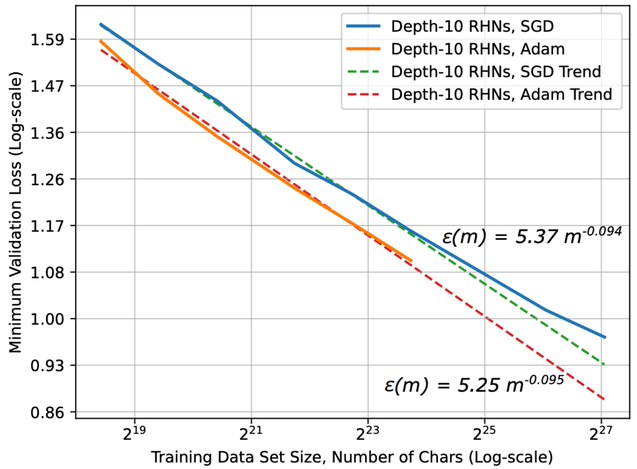
- General power law: $y = Ax^{\alpha}$

OpenAI (Kaplan et al. 2020)

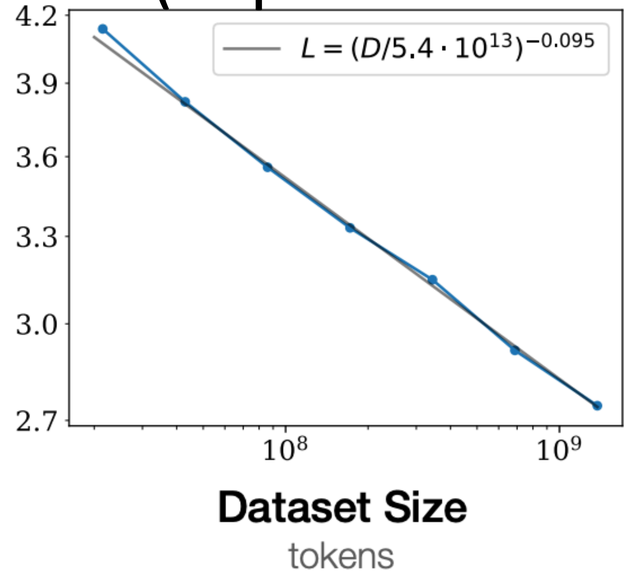


Dataset Scaling Laws

Baidu (Hestness et al. 2017)



OpenAI (Kaplan et al. 2020)

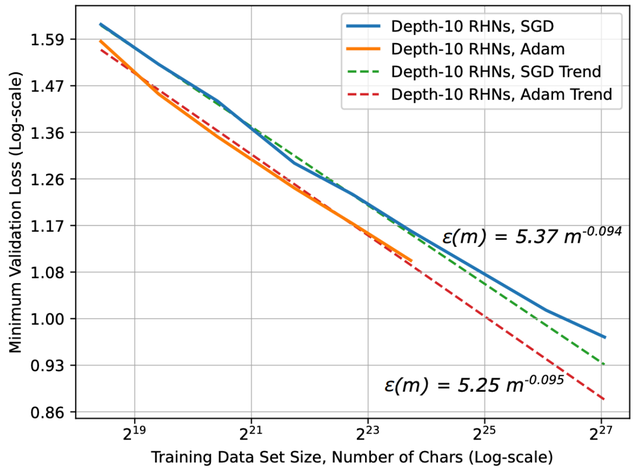


- General power law: $y = Ax^{\alpha}$
- Dataset scaling law:

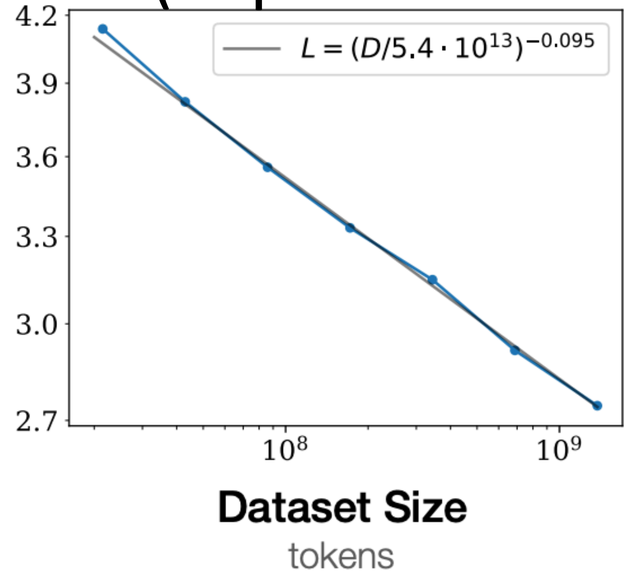
$$L(D) \approx \left(\frac{D}{D_c}\right)^{-\alpha_D}$$

Dataset Scaling Laws

Baidu (Hestness et al. 2017)



OpenAI (Kaplan et al. 2020)



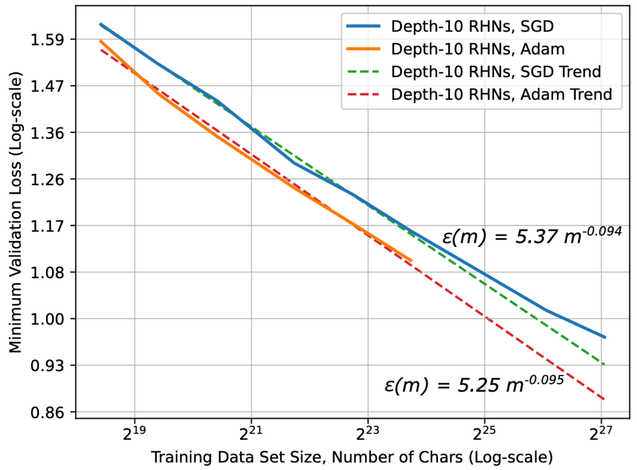
- General power law: $y = Ax^{\alpha}$
- Dataset scaling law:

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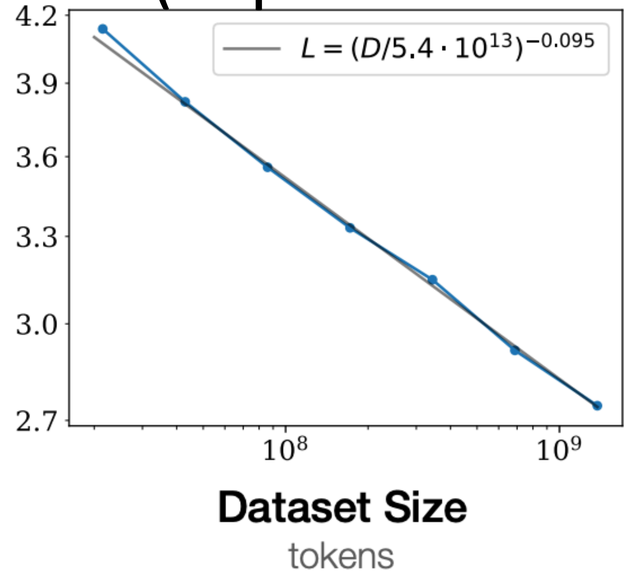
$$= \underbrace{D_c^{\alpha_D}}_{A} \underbrace{D^{-\alpha_D}}_{x^{\alpha}}$$

Dataset Scaling Laws

Baidu (Hestness et al. 2017)



OpenAI (Kaplan et al. 2020)



- General power law: $y = Ax^{\alpha}$
- Dataset scaling law:

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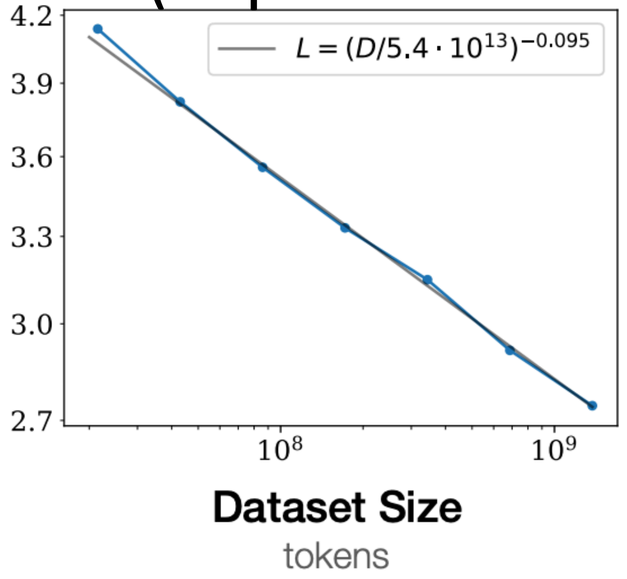
$$= \underbrace{D_c^{\alpha_D}}_{A} \underbrace{D^{-\alpha_D}}_{x^{\alpha}}$$

$$\propto D^{-\alpha_D}$$

Dataset Scaling Laws

- The exponent captures the fundamental scaling behavior:
  i.e. how quickly loss decreases as D increases?

OpenAI (Kaplan et al. 2020)

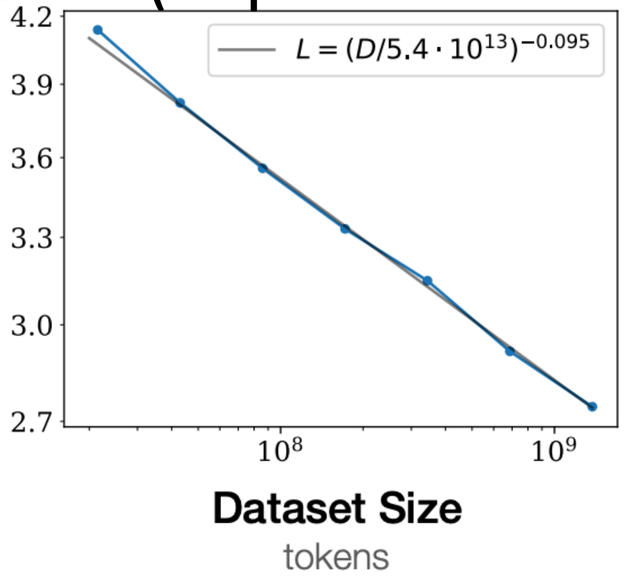


Dataset Scaling Laws

- The exponent captures the fundamental scaling behavior:
  i.e. how quickly loss decreases as D increases?

$$L(D) = \left(\frac{D_c}{D}\right)^{\alpha_D} = \left(\frac{D}{D_c}\right)^{-\alpha_D} = D_c^{\alpha_D} D^{-\alpha_D} \propto D^{-\alpha_D}$$

OpenAI (Kaplan et al. 2020)

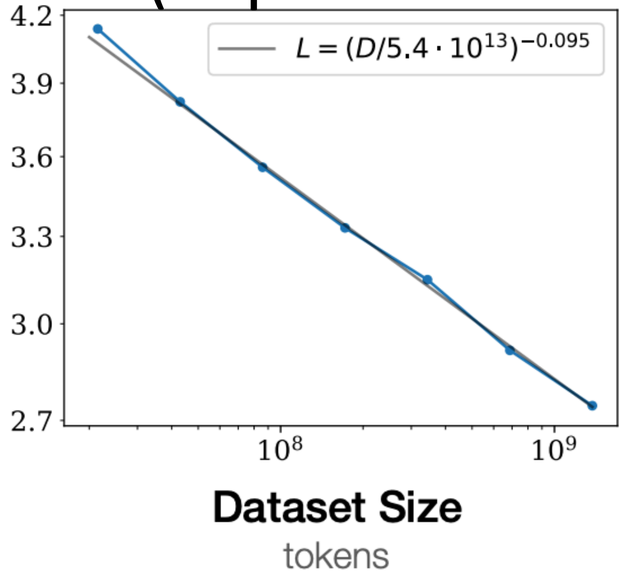


Dataset Scaling Laws

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OpenAI (Kaplan et al. 2020)



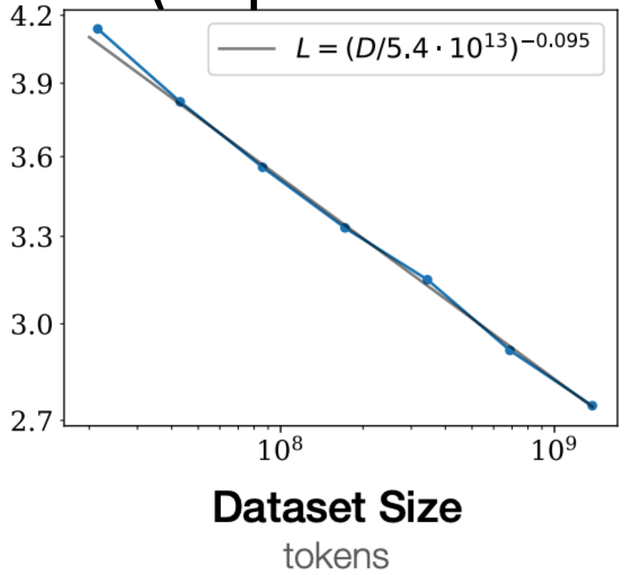
- For example, $L(D) \propto D^{-0.095}$

Dataset Scaling Laws

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  i.e. how quickly loss decreases as D increases?

$$L(D) = \left(\frac{D_c}{D}\right)^{\alpha_D} = \left(\frac{D}{D_c}\right)^{-\alpha_D} = D_c^{\alpha_D} D^{-\alpha_D} \propto D^{-\alpha_D}$$

OpenAI (Kaplan et al. 2020)

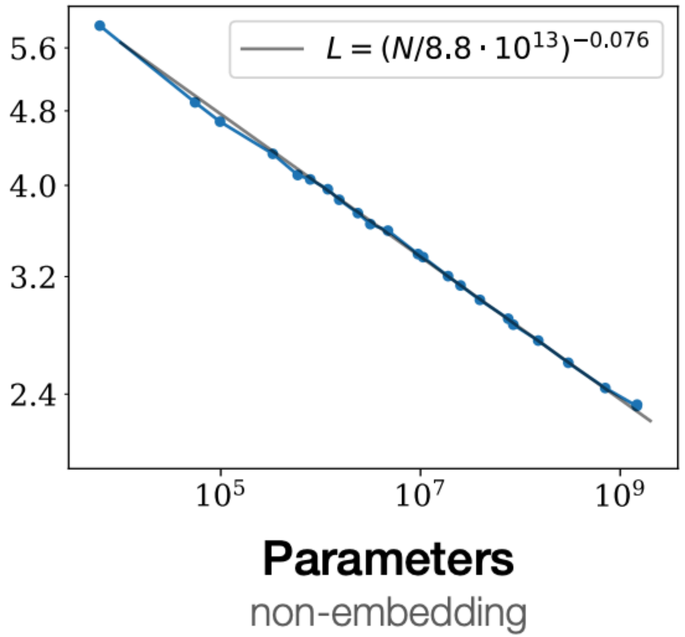


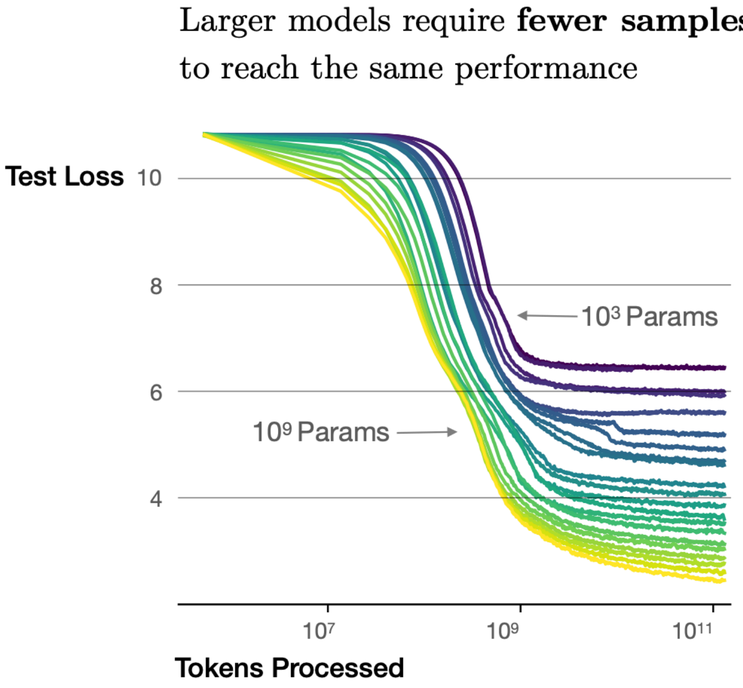
- For example, $L(D) \propto D^{-0.095}$
- Increasing D by 10x, will reduce loss by ~20%

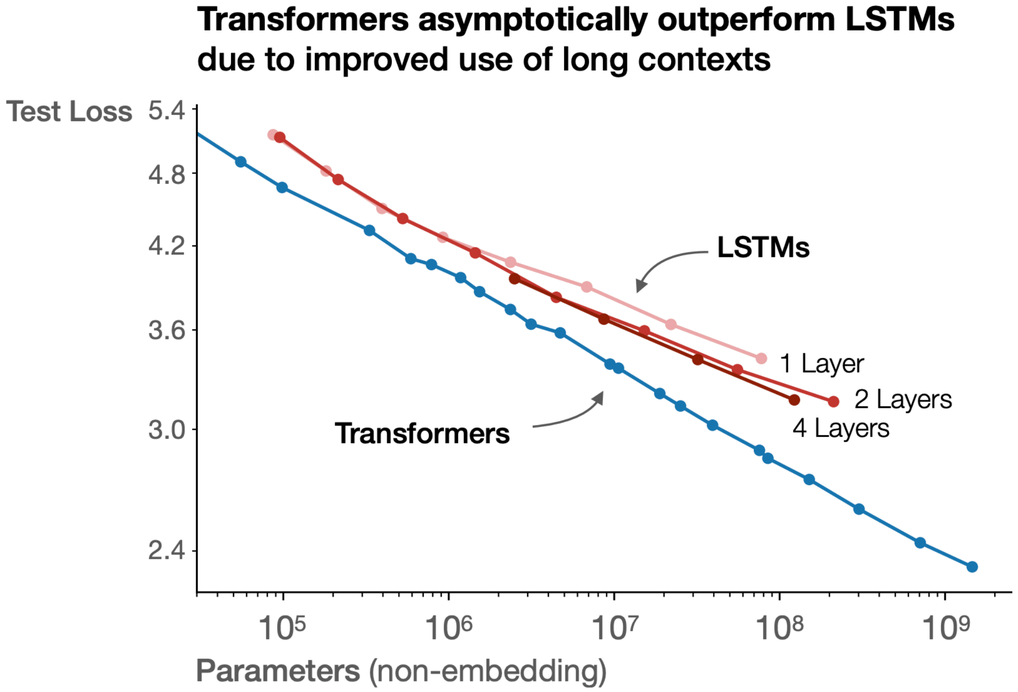
$$\frac{L(10D)}{L(D)} = \frac{(10D)^{-0.095}}{D^{-0.095}} = 10^{-0.095} \approx 0.803$$

Model Size Scaling Laws

- Similar to the dataset scaling laws

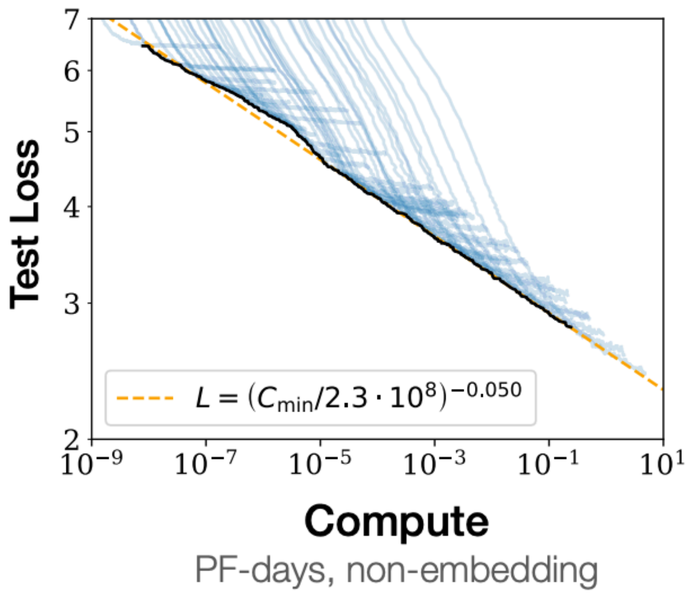






OpenAI (Kaplan et al. 2020)

Compute Scaling Laws

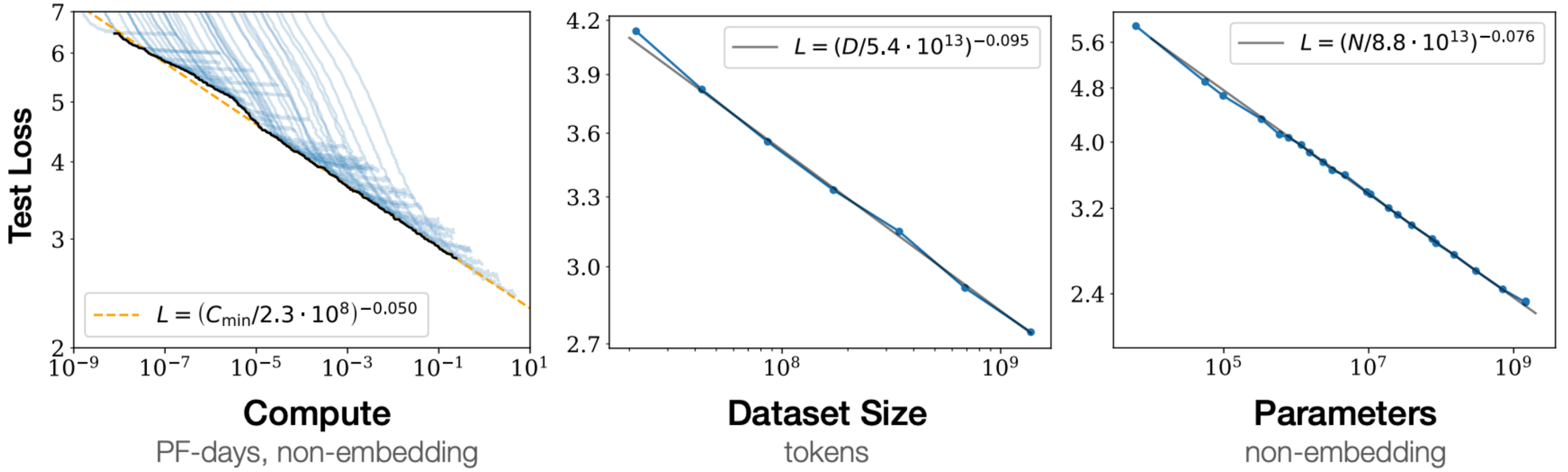


- Blue lines: model of different sizes N, trained on increasing amount of data D
- **Black** line - “compute-efficient frontier”: lowest test loss achieved by any model at a given compute budget
- Orange dashed line: use log-space regression to fit a power law

OpenAI (Kaplan et al. 2020)

GPT Scaling Laws

- With more compute, larger models get further down the loss “frontier”
- Building a bigger model (increasing compute) will decrease test loss!



$$L(C) \propto C^{-0.050}$$

$$L(D) \propto D^{-0.095}$$

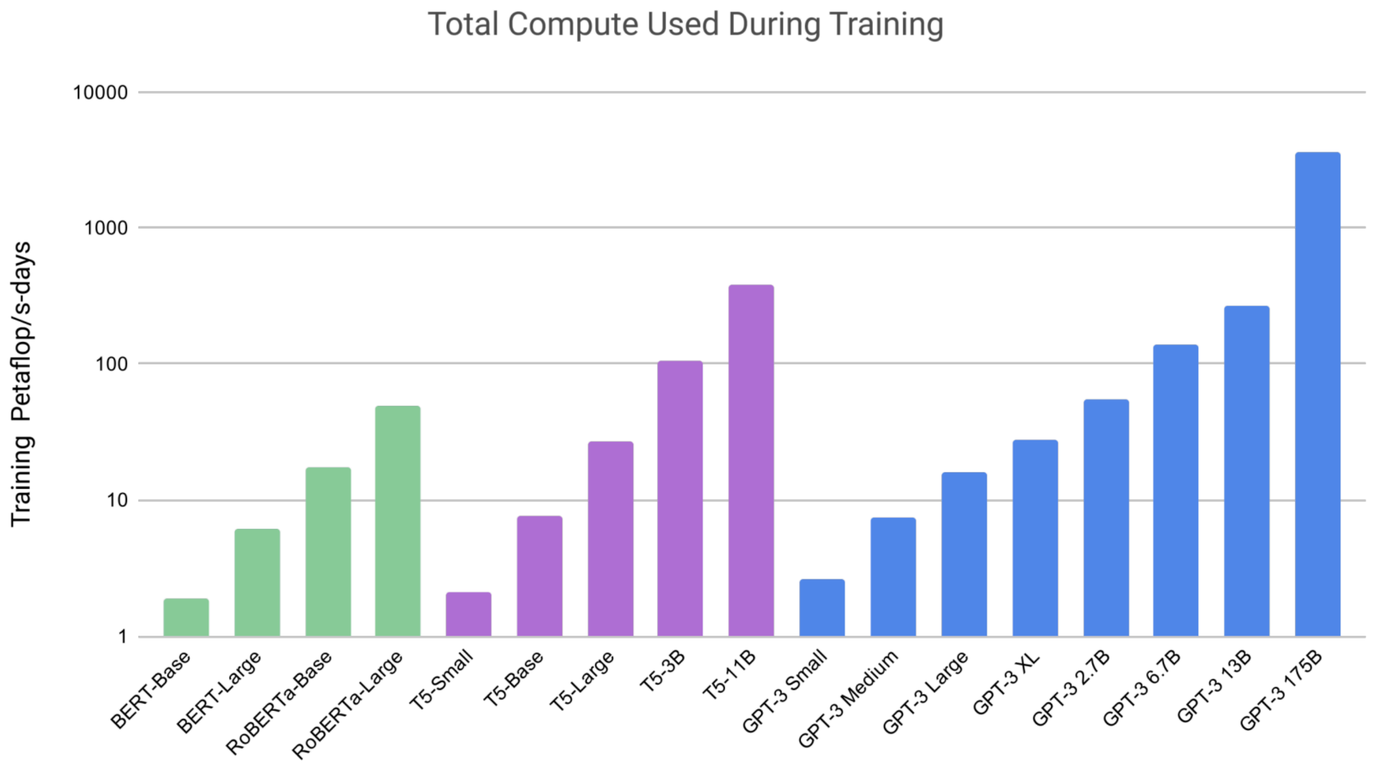
$$L(N) \propto N^{-0.076}$$

Kaplan et al. (2020)

GPT Scaling Laws

- For a fixed compute budget, Kaplan et al. (2020) argued for scaling model size much faster than training data (train bigger models!):

$$N_{\text{opt}} \propto C^{0.73}, \qquad D_{\text{opt}} \propto C^{0.27}$$

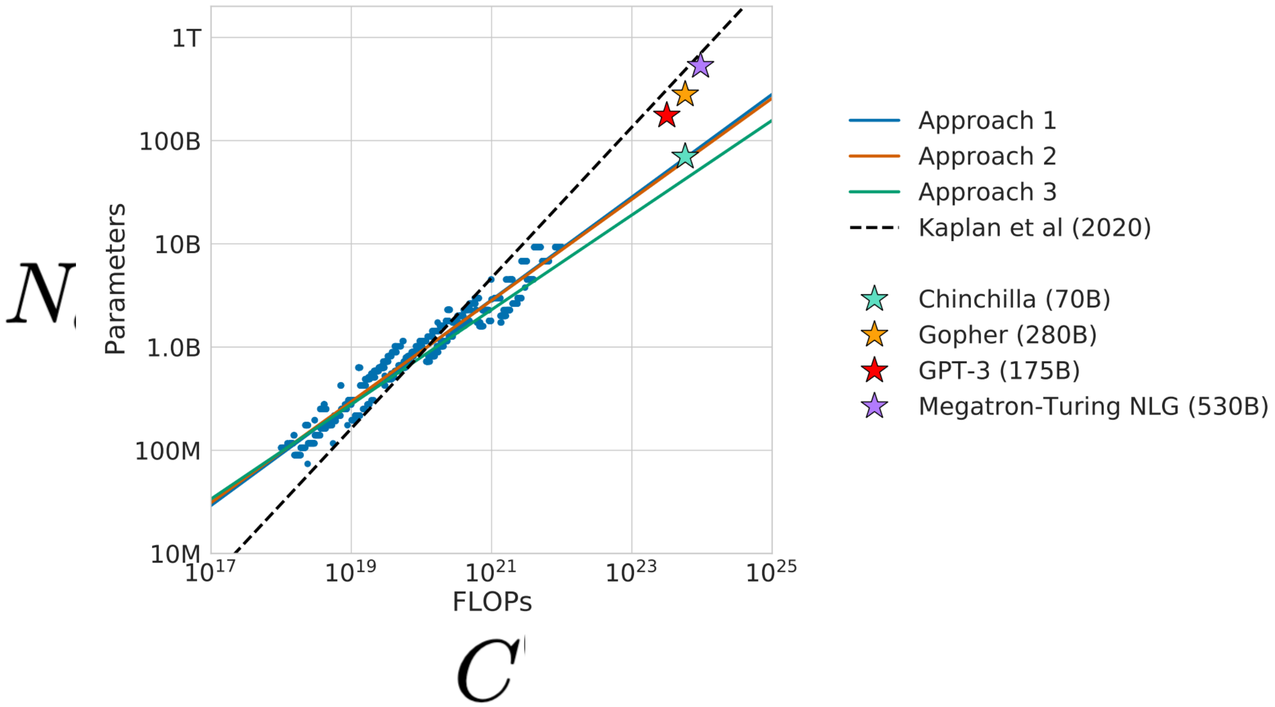


Kaplan et al. (2020)

Chinchilla Scaling Laws

- In contrast, Hoffmann (2022) argued for scaling model size and training data equally (training models on more data, avoid under-training):

$$N_{\text{opt}} \propto C^{0.50}, \qquad D_{\text{opt}} \propto C^{0.50}$$



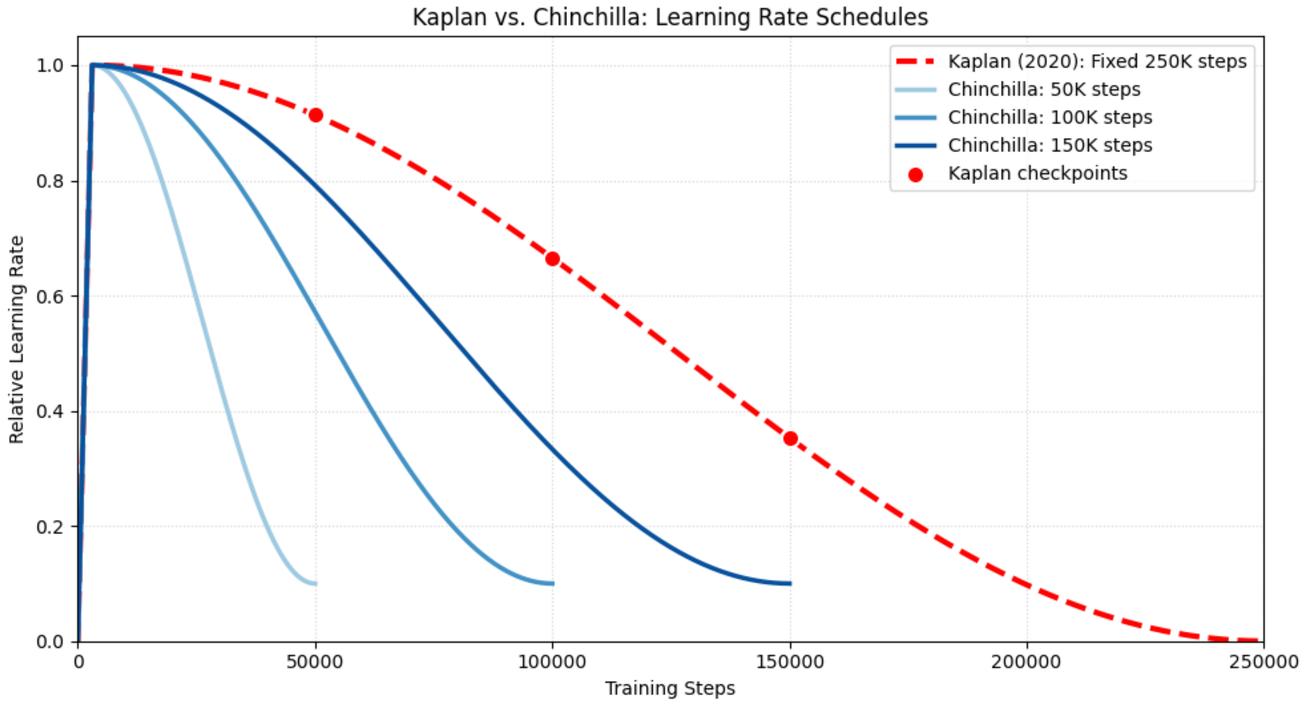
MMLU results:

Random	25.0%
Average human rater	34.5%
GPT-3 5-shot	43.9%
Gopher 5-shot	60.0%
***Chinchilla* 5-shot**	**67.6%**
Average human expert performance	*89.8%*

Hoffmann et al. (2022)

Chinchilla Scaling Laws

- Learning rate: adaptive schedule vs. Kaplan+ (2020)'s fixed schedule
- Model size: up to 70B vs. Kaplan+ (2020)'s 1.5B non-embedding parameters



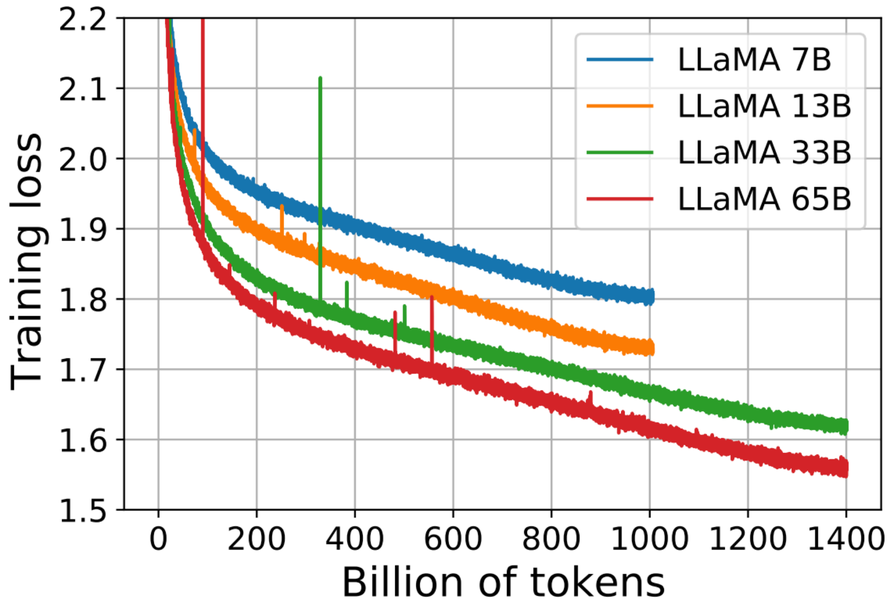
Hoffmann et al. (2022)

Llama (Preview)

- Besides scale dataset and model sizes for **training** compute budget, it is also important to consider the **inference (test-time)** compute budget

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

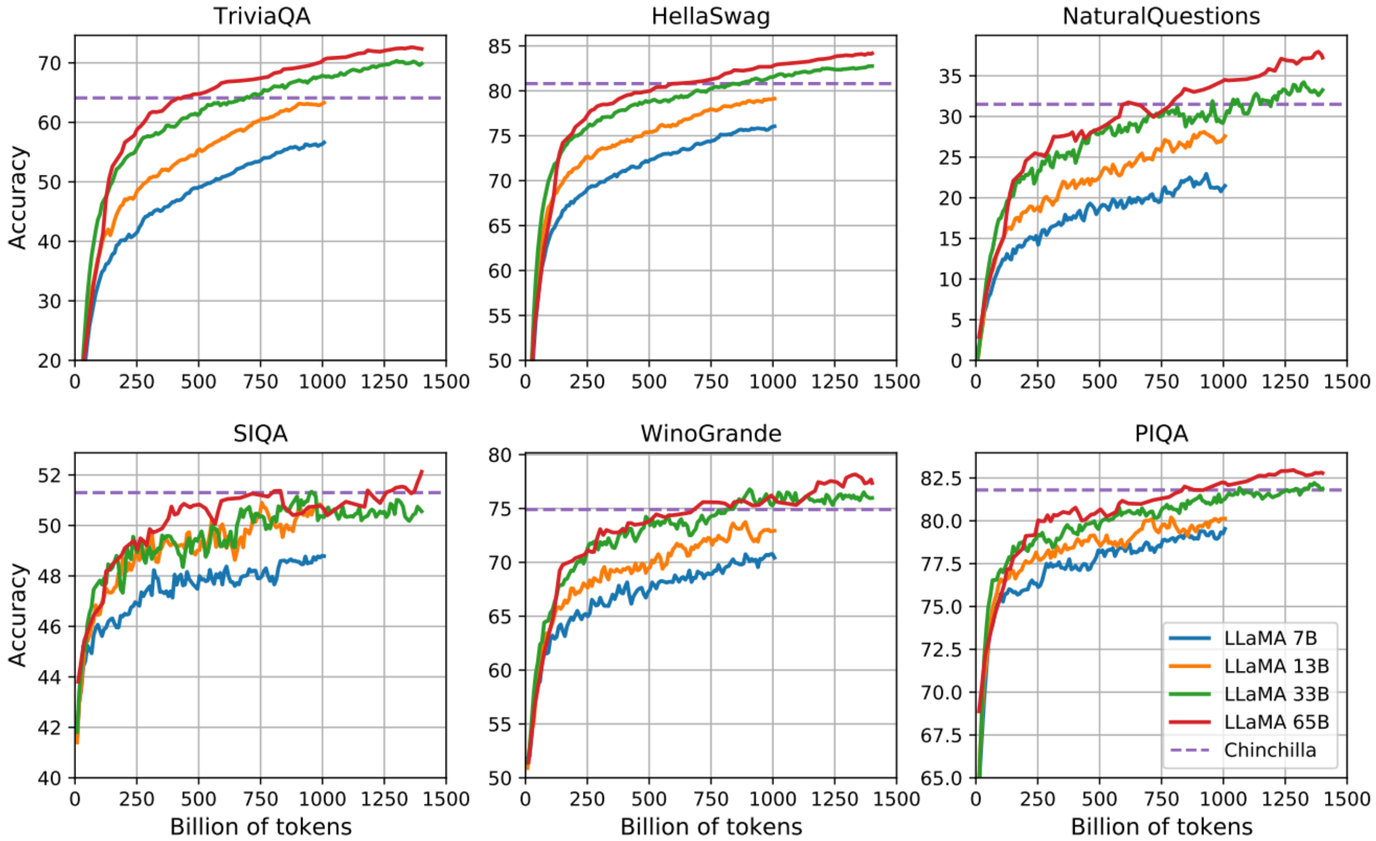
trained on 1~1.4T tokens



Touvron et al. (2023)

Llama (Preview)

- Smaller models continue to improve when trained longer on substantially more tokens.



Touvron et al. (2023)

Baidu Scaling Laws

- Scaling behavior wasn't unique to Transformers or LLMs.
- Irreducible error:

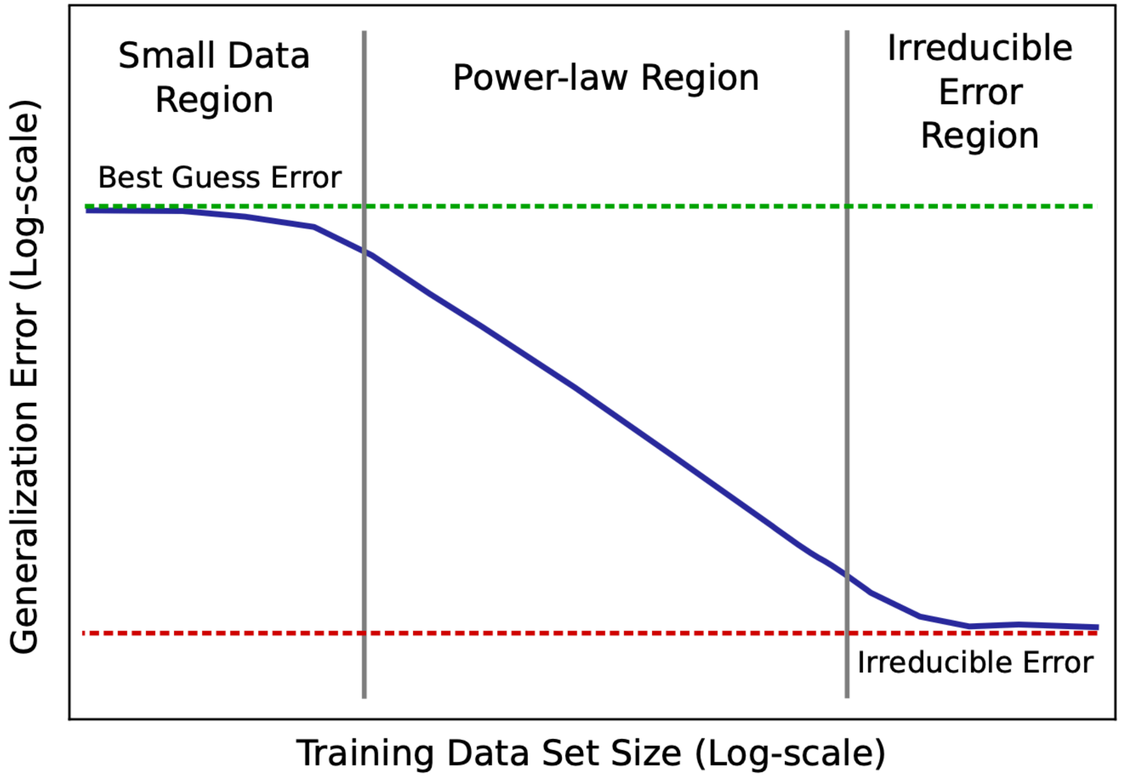
$$L(D) = L_\infty + AD^{-\alpha_D}$$

- e.g., cross-entropy loss in LM:

$$H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$$

true predicted

"I want a cup of ____"



Hestness et al. (2017)

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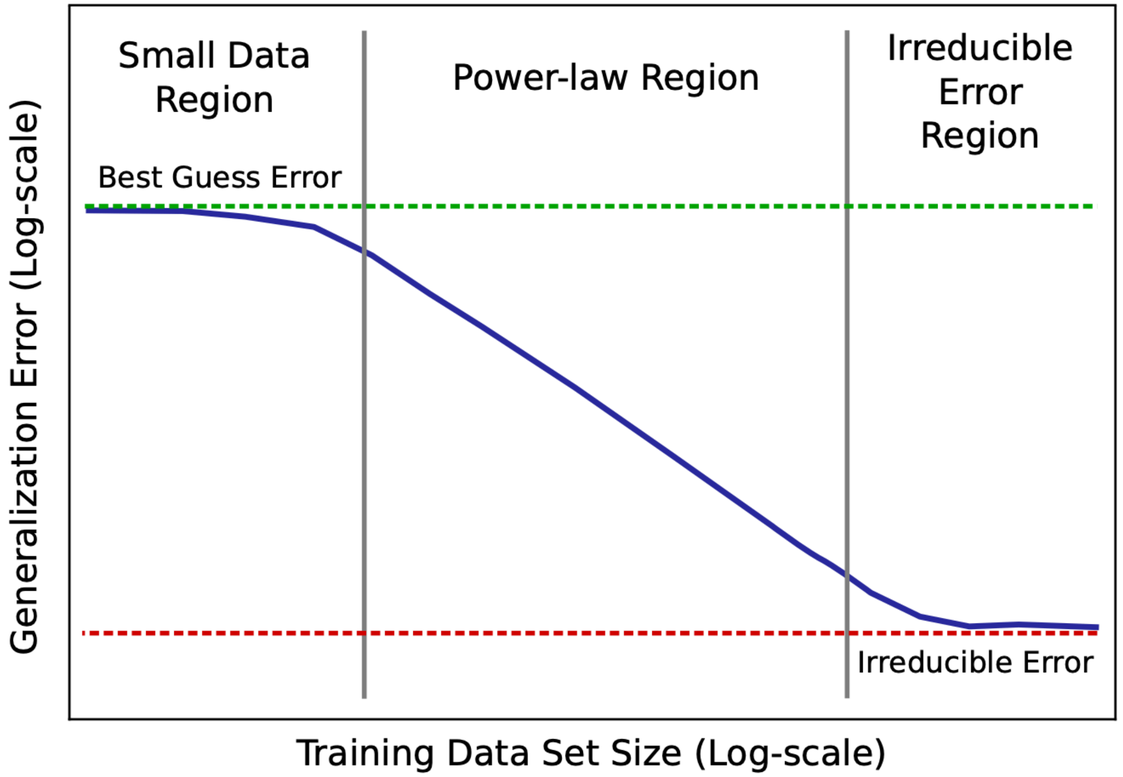
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Hestness et al. (2017)

Summary: Scaling Laws

- Hestness+ 2017: Scaling behavior wasn't unique to Transformers or LLMs.
- Kaplan+ 2020 (GPT-3): Prioritize model size, make models bigger
- Hoffmann+ 2022 (Chinchilla): Increase training data & model size equally
- Touvron+ 2023 (Llama-1): Smaller models continue to improve when trained on substantially more data, benefiting inference efficiency.

- Later: data-constrained scaling, sparse/MoE, test-time scaling, …
- And more: synthetic data, code, multimodal data, data filtering/curation, repeated data, post-training, reinforcement learning